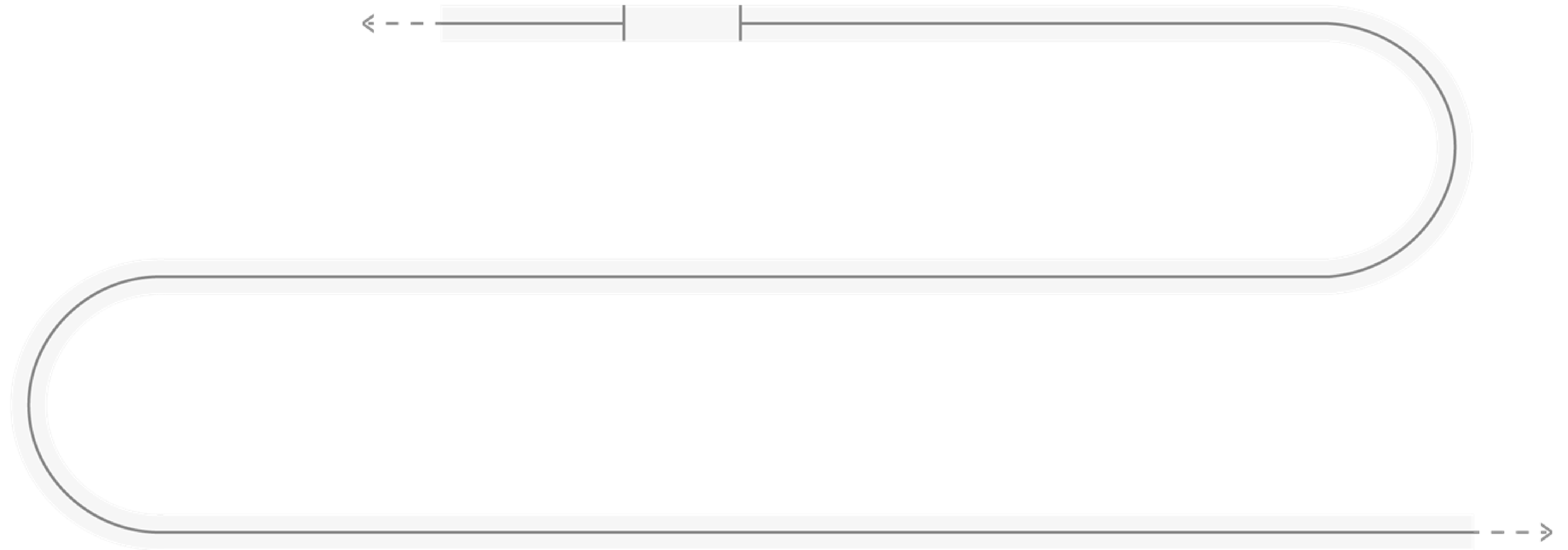


Neuro e IA: coevolución



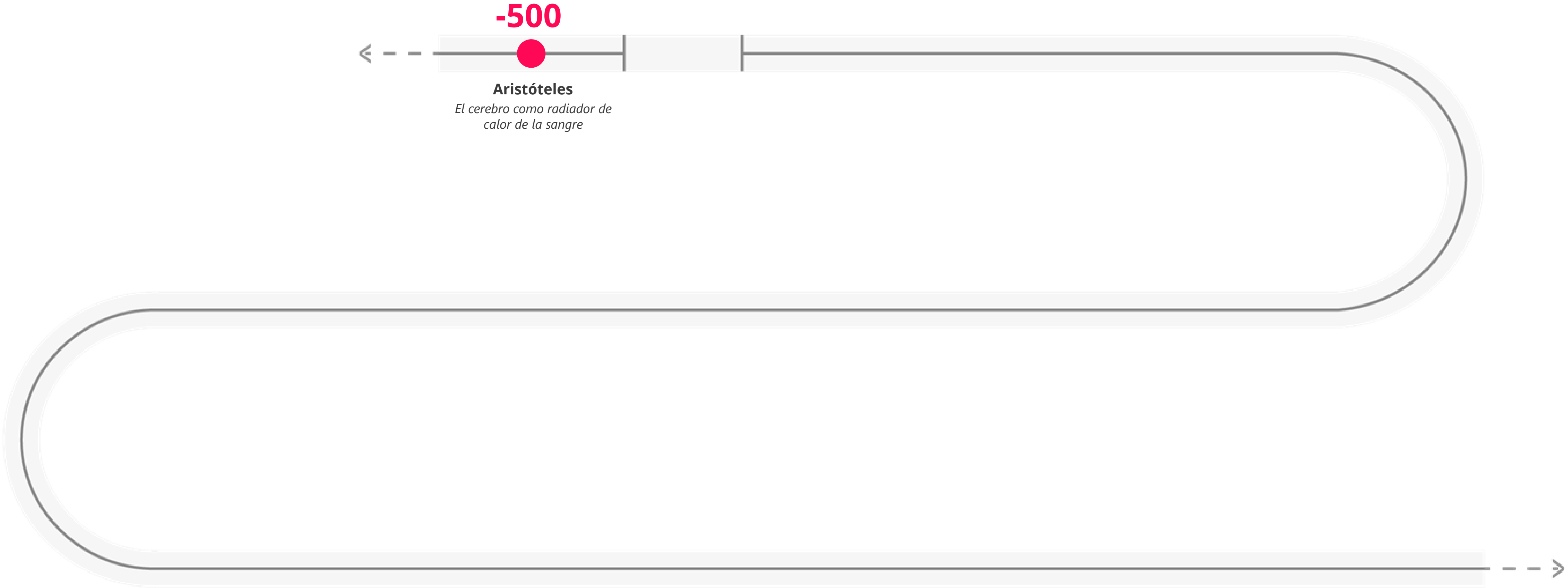


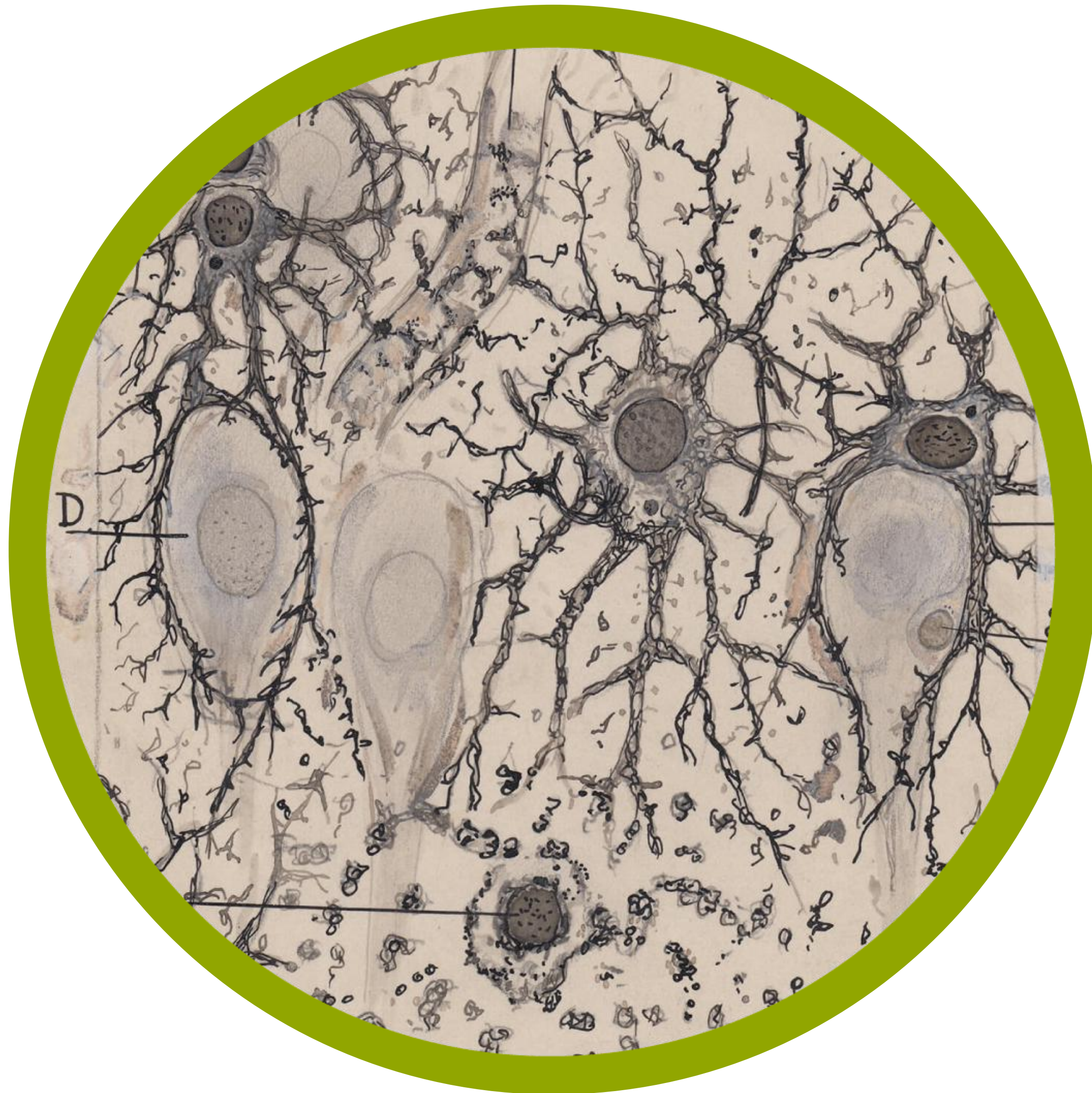
-500

Aristóteles

*El cerebro como radiador
de calor de la sangre*

Neuro e IA: coevolución



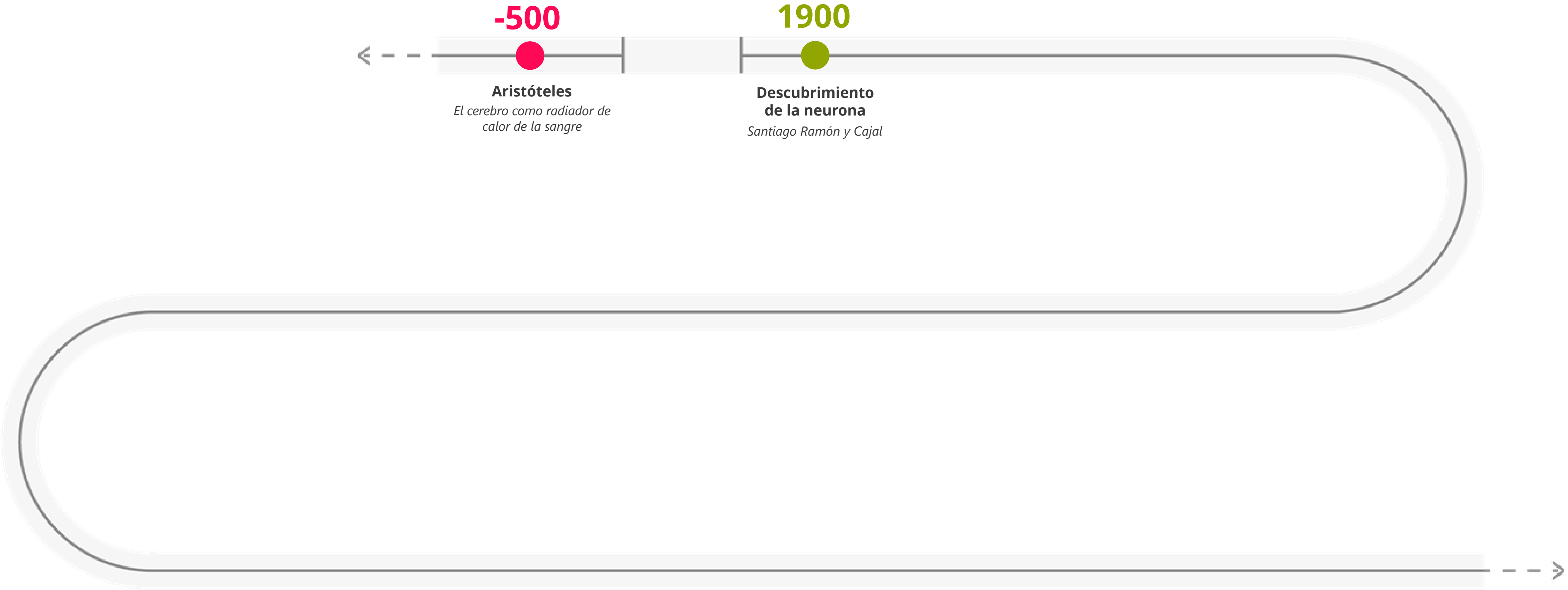


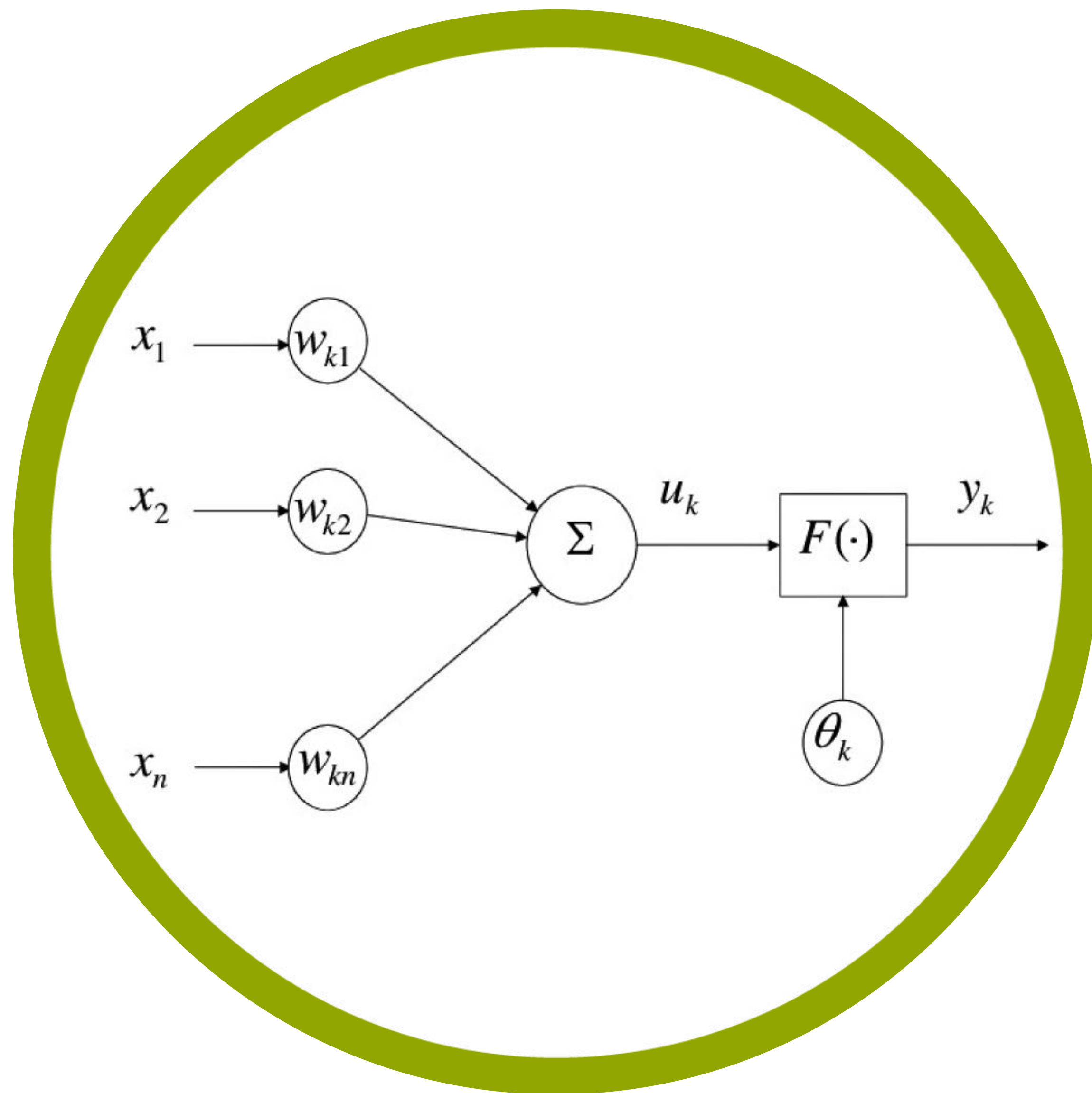
1900

Descubrimiento de la neurona

Santiago Ramón y Cajal

Neuro e IA: coevolución



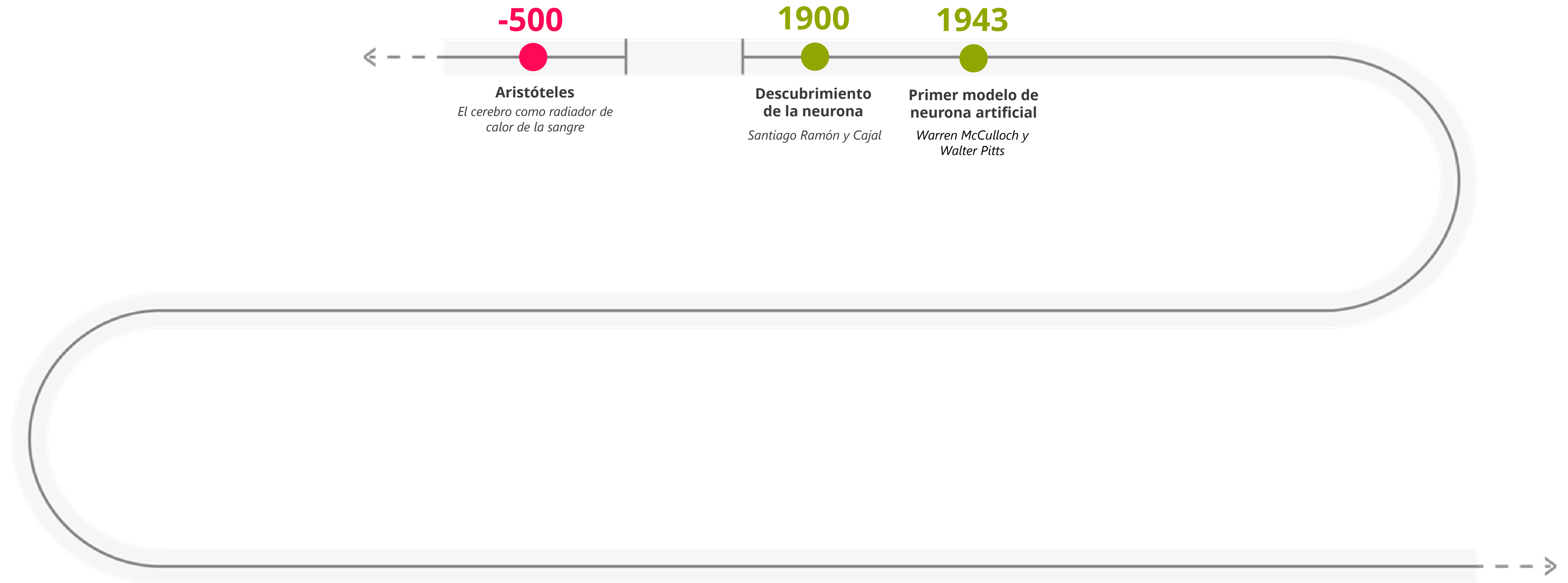


1943

Primer modelo de neurona artificial

Warren McCulloch y Walter Pitts

Neuro e IA: coevolución





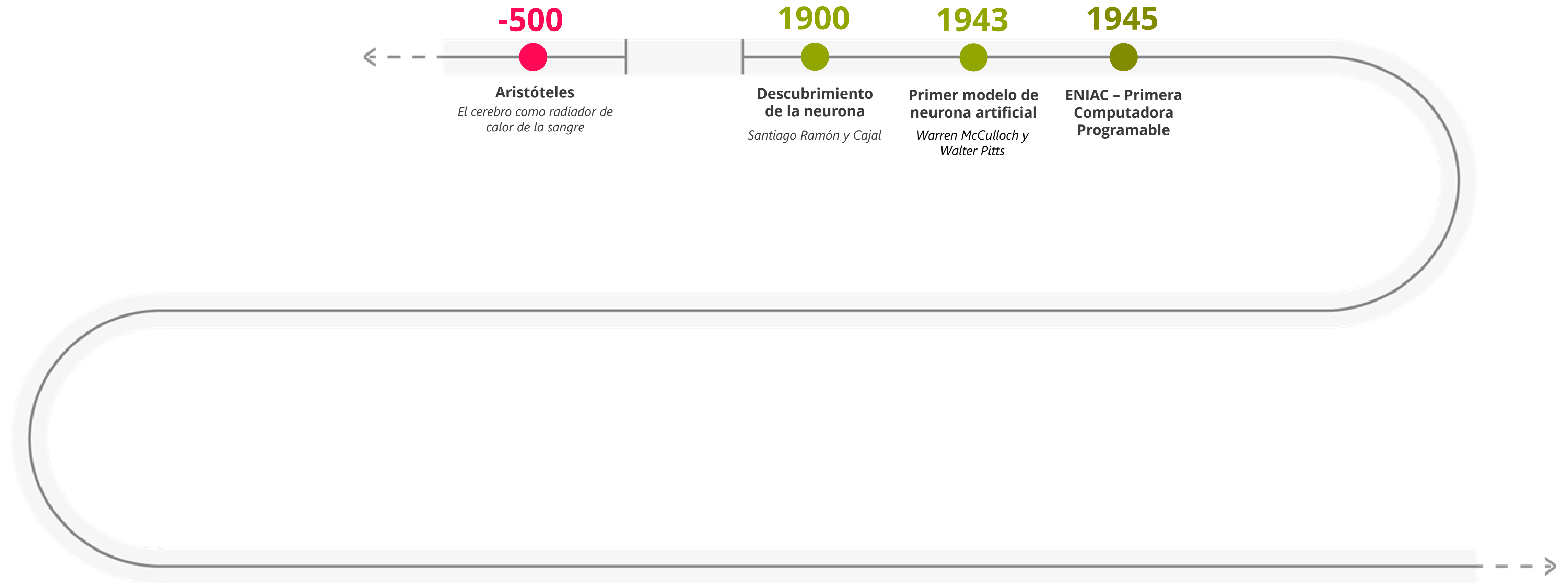
1945

ENIAC – Primera Computadora Programable

Conceptos computacionales

- *Información*
- *Algoritmo*
- *Memoria de corto y largo plazo*
- *Sistema de procesamiento de la información*
- *Distinción entre Software y Hardware*

Neuro e IA: coevolución





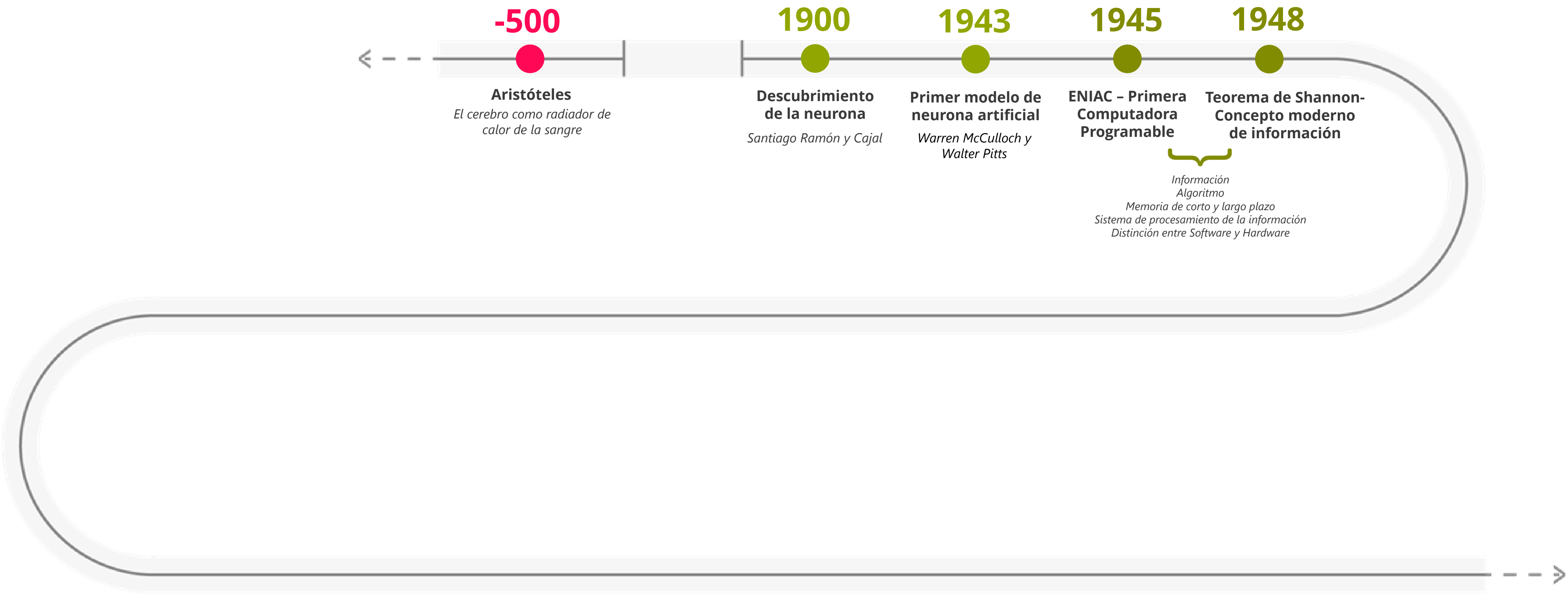
1948

Teorema de Shannon- Concepto moderno de información

Conceptos computacionales

- *Información*
- *Algoritmo*
- *Memoria de corto y largo plazo*
- *Sistema de procesamiento de la información*
- *Distinción entre Software y Hardware*

Neuro e IA: coevolución



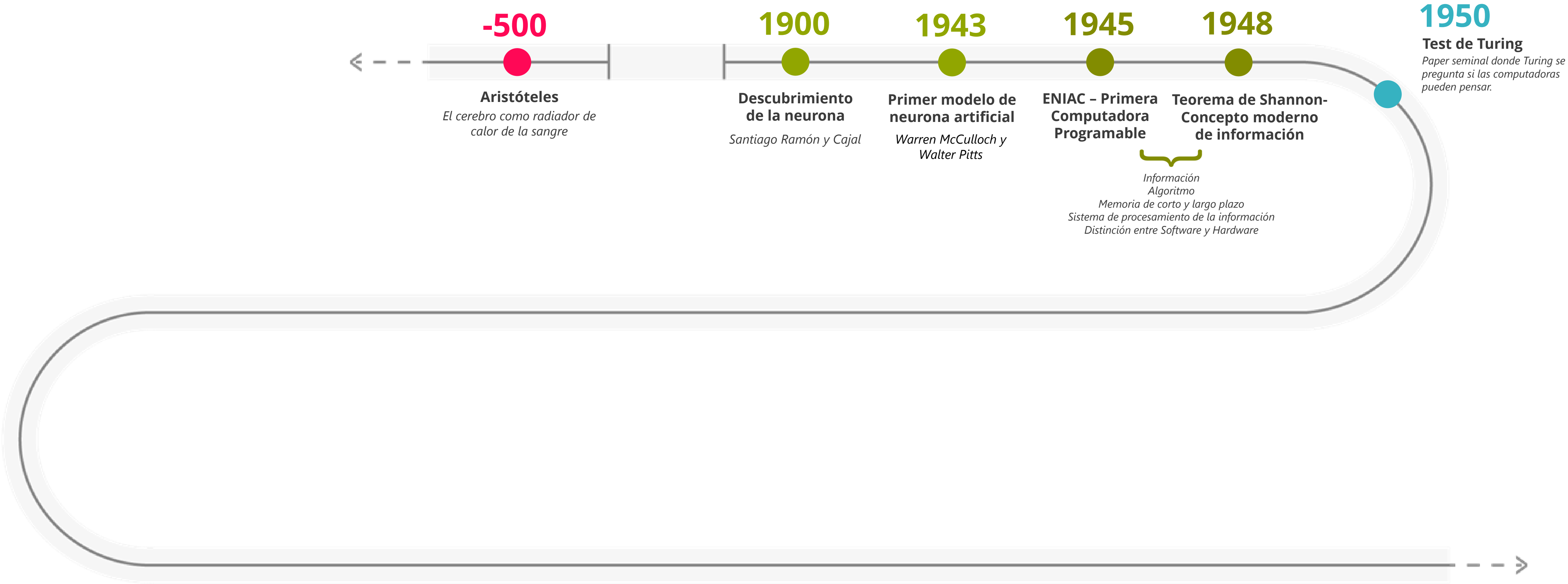


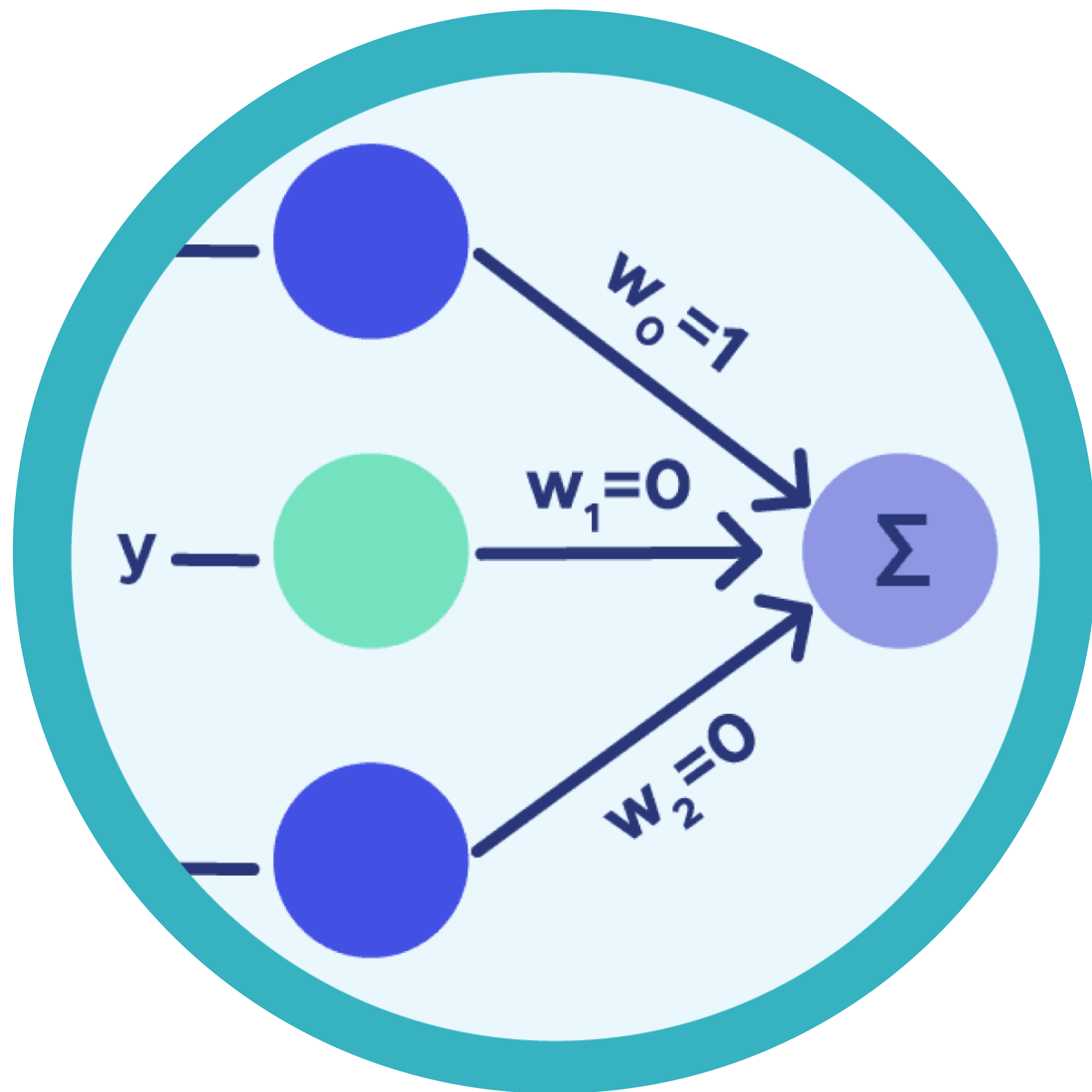
1950

Test de Turing

Paper seminal donde Turing se pregunta si las computadoras pueden pensar.

Neuro e IA: coevolución



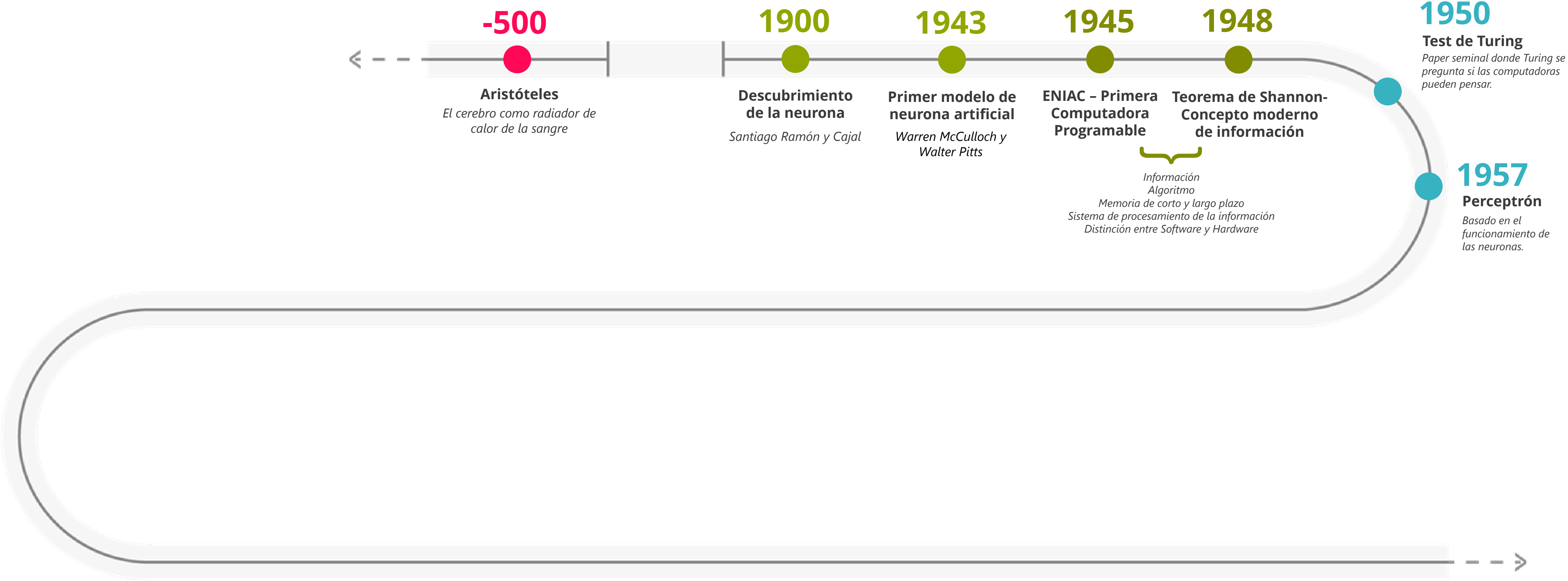


1957

Perceptrón

Basado en el funcionamiento de las neuronas.

Neuro e IA: coevolución



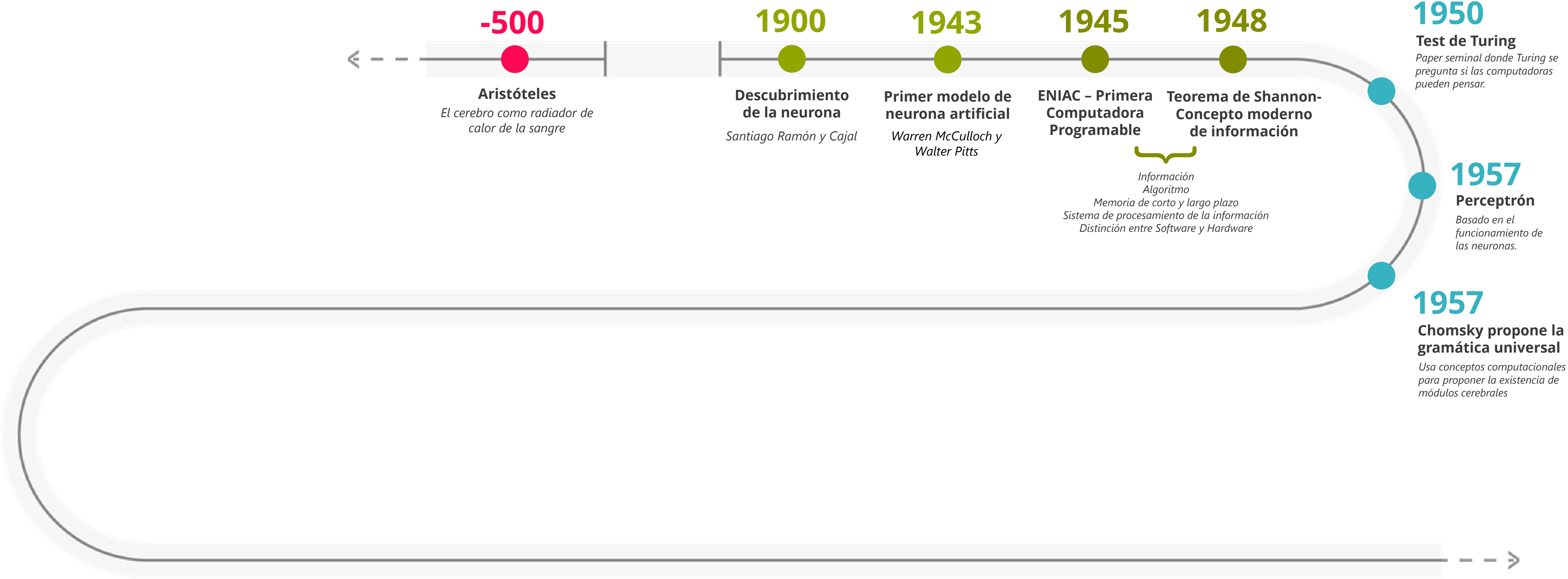


1957

Chomsky propone la gramática universal

Usa conceptos computacionales para proponer la existencia de módulos cerebrales

Neuro e IA: coevolución

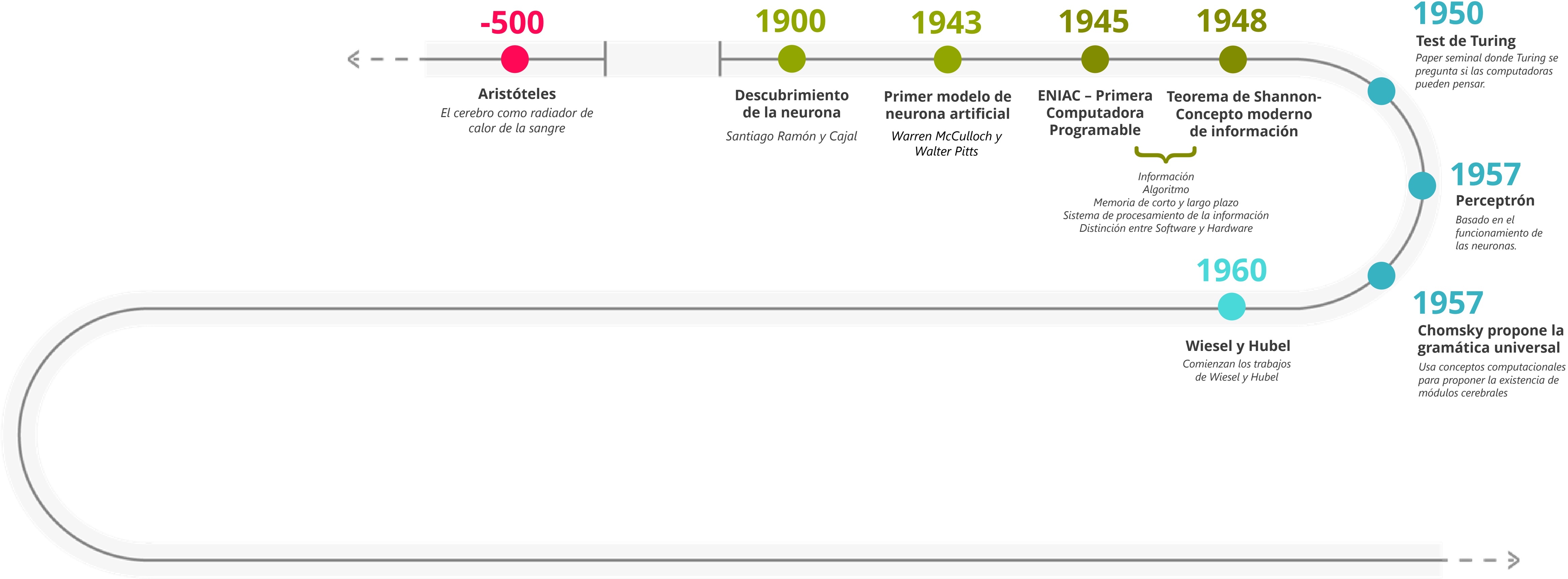




1960

**Comienzan los trabajos de
Wiesel y Hubel**

Neuro e IA: coevolución

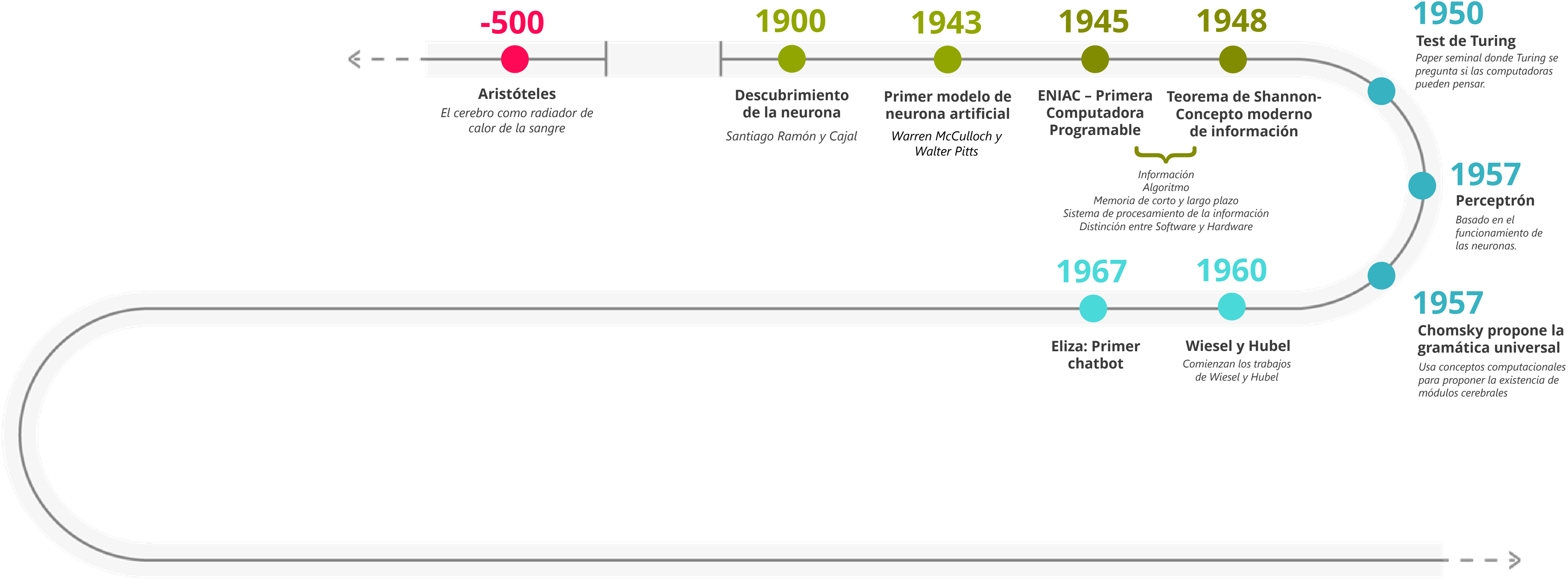




1967

Eliza: Primer chatbot

Neuro e IA: coevolución



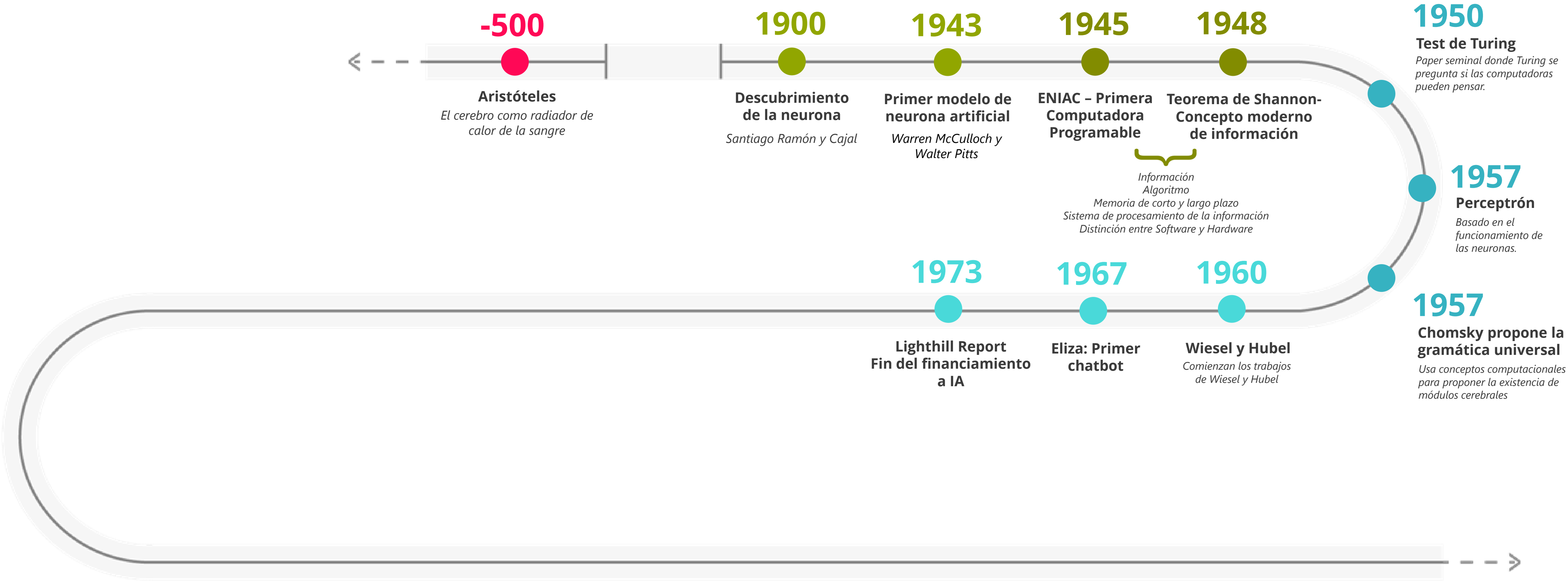


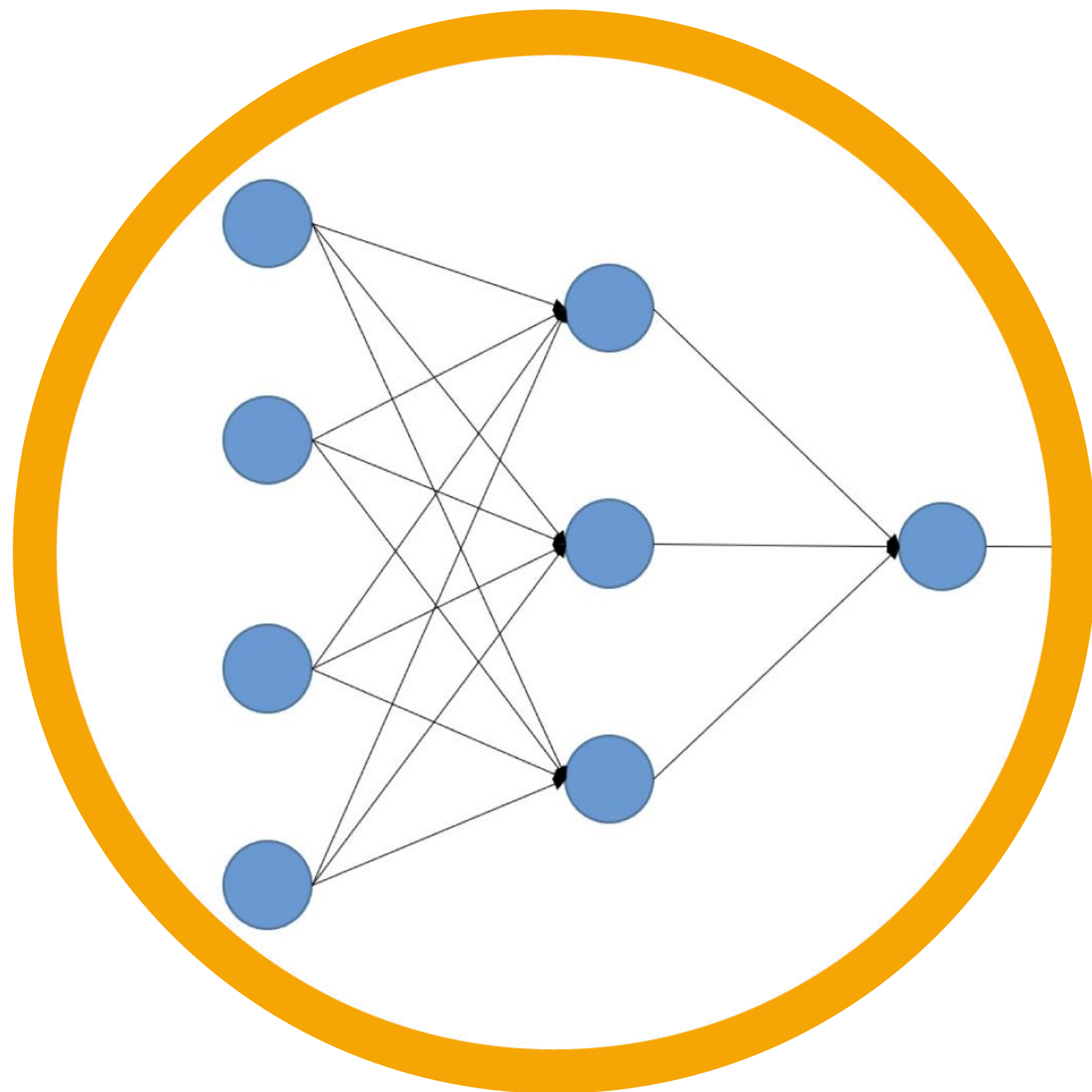
1973

Lighthill Report

***Fin del financiamiento de la
Inteligencia Artificial***

Neuro e IA: coevolución

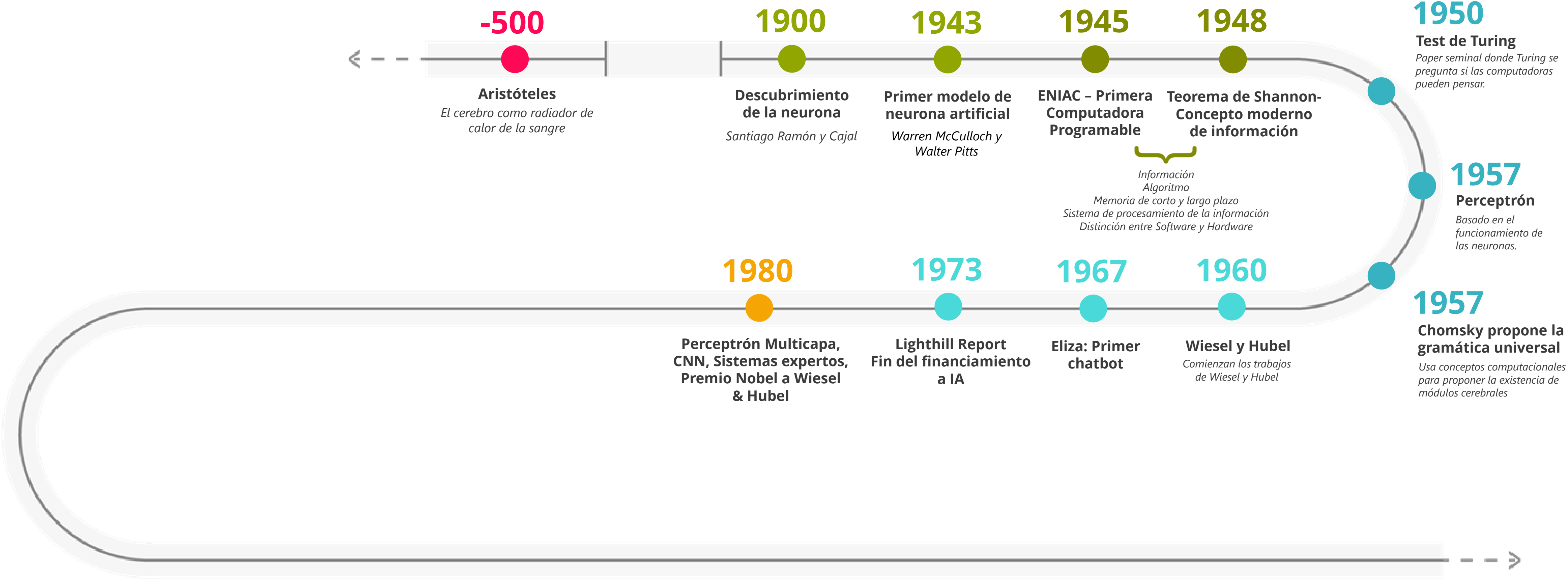




1980

**Perceptrón Multicapa,
CNN, Sistemas expertos,
Premio Nobel a Wiesel
& Hubel**

Neuro e IA: coevolución



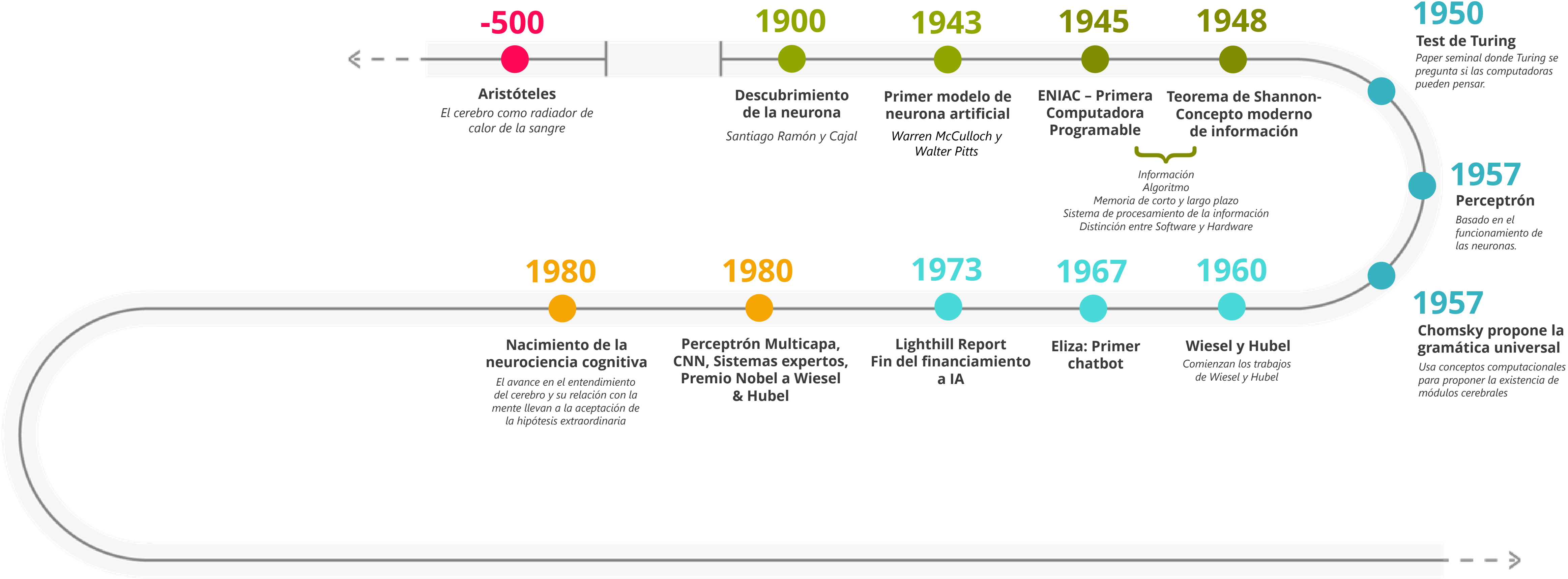


1980

Nacimiento de la Neurociencia Cognitiva

*El avance en el entendimiento
del cerebro y su relación con la
mente llevan a la aceptación de
la hipótesis extraordinaria*

Neuro e IA: coevolución

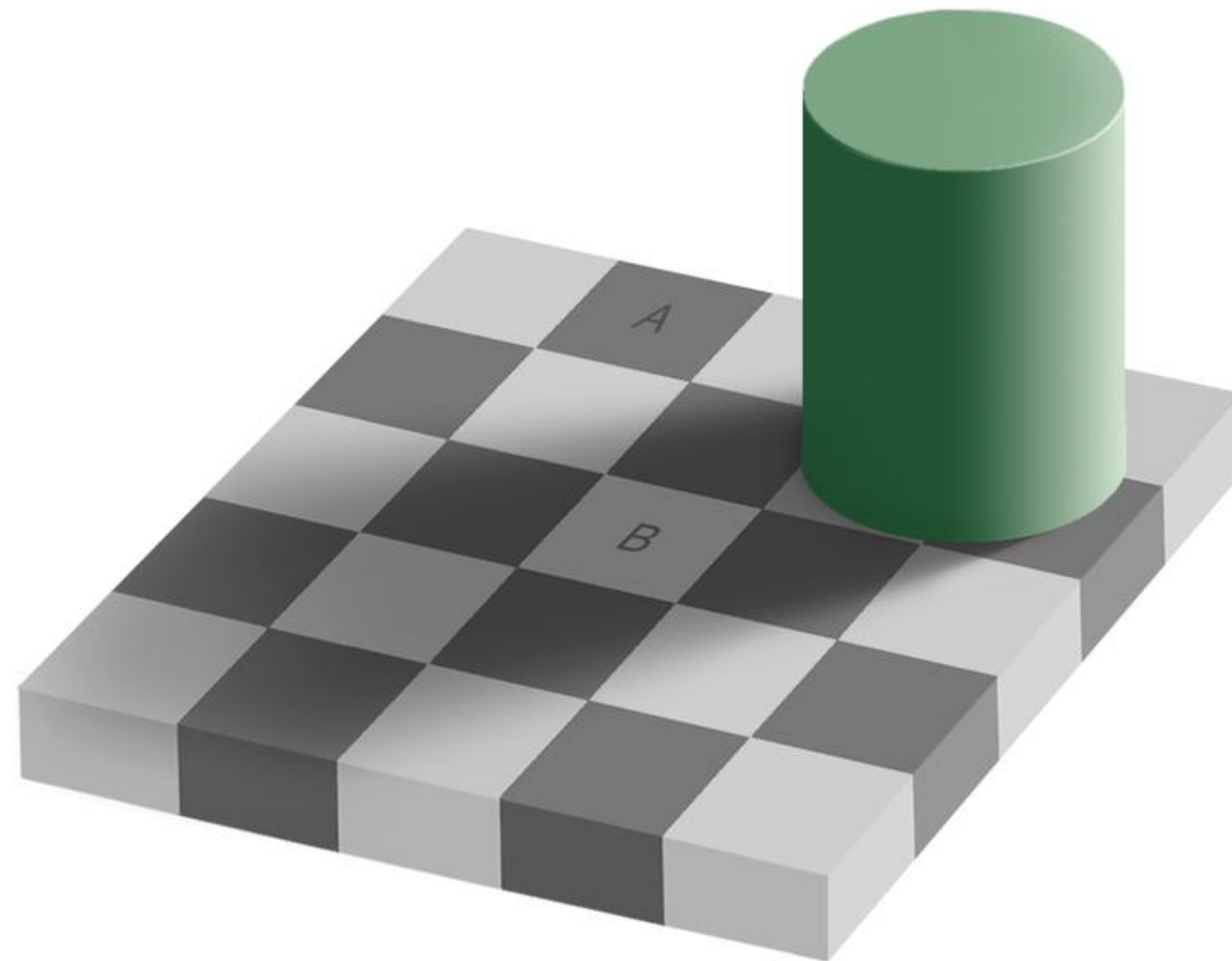


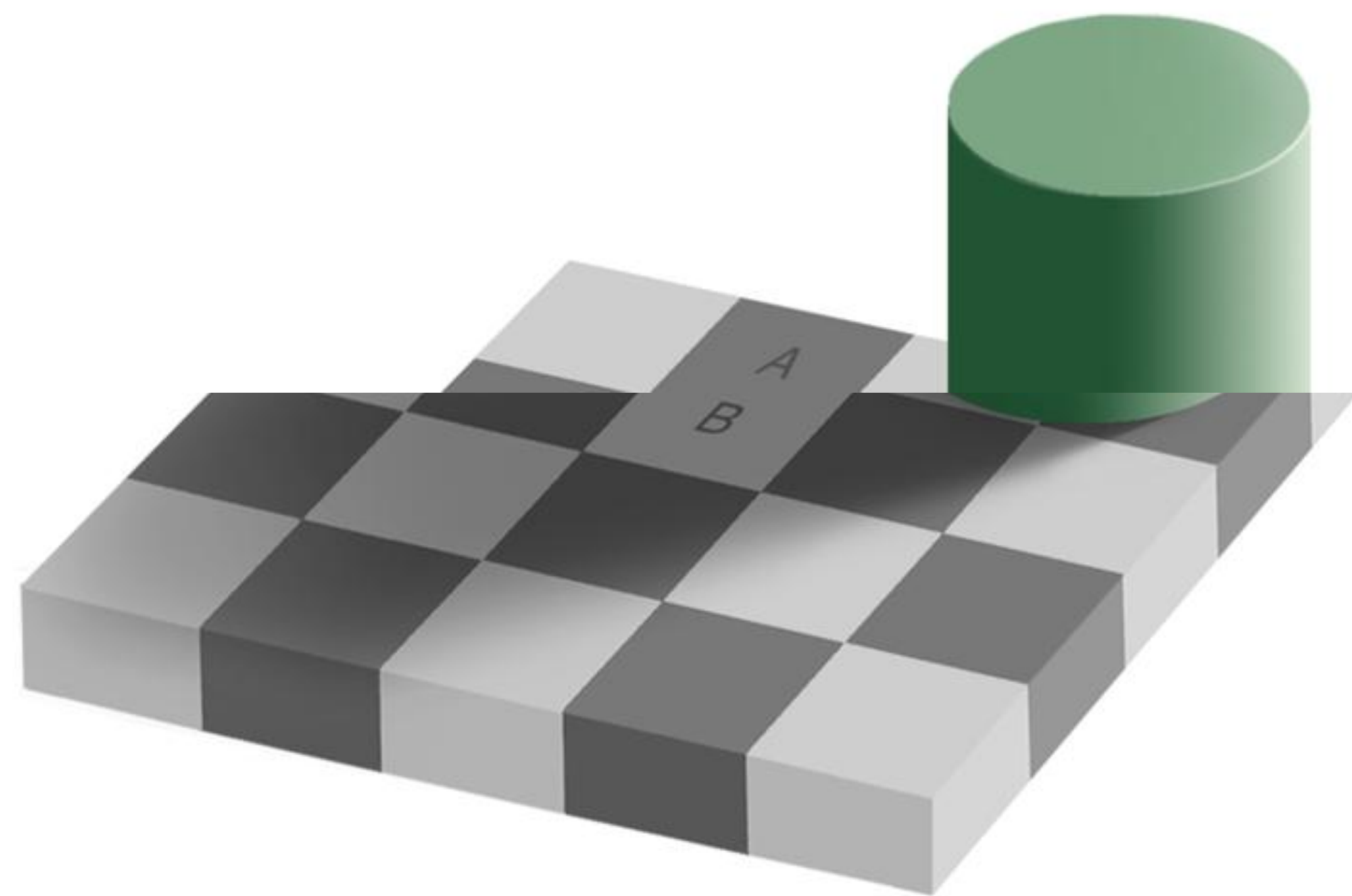


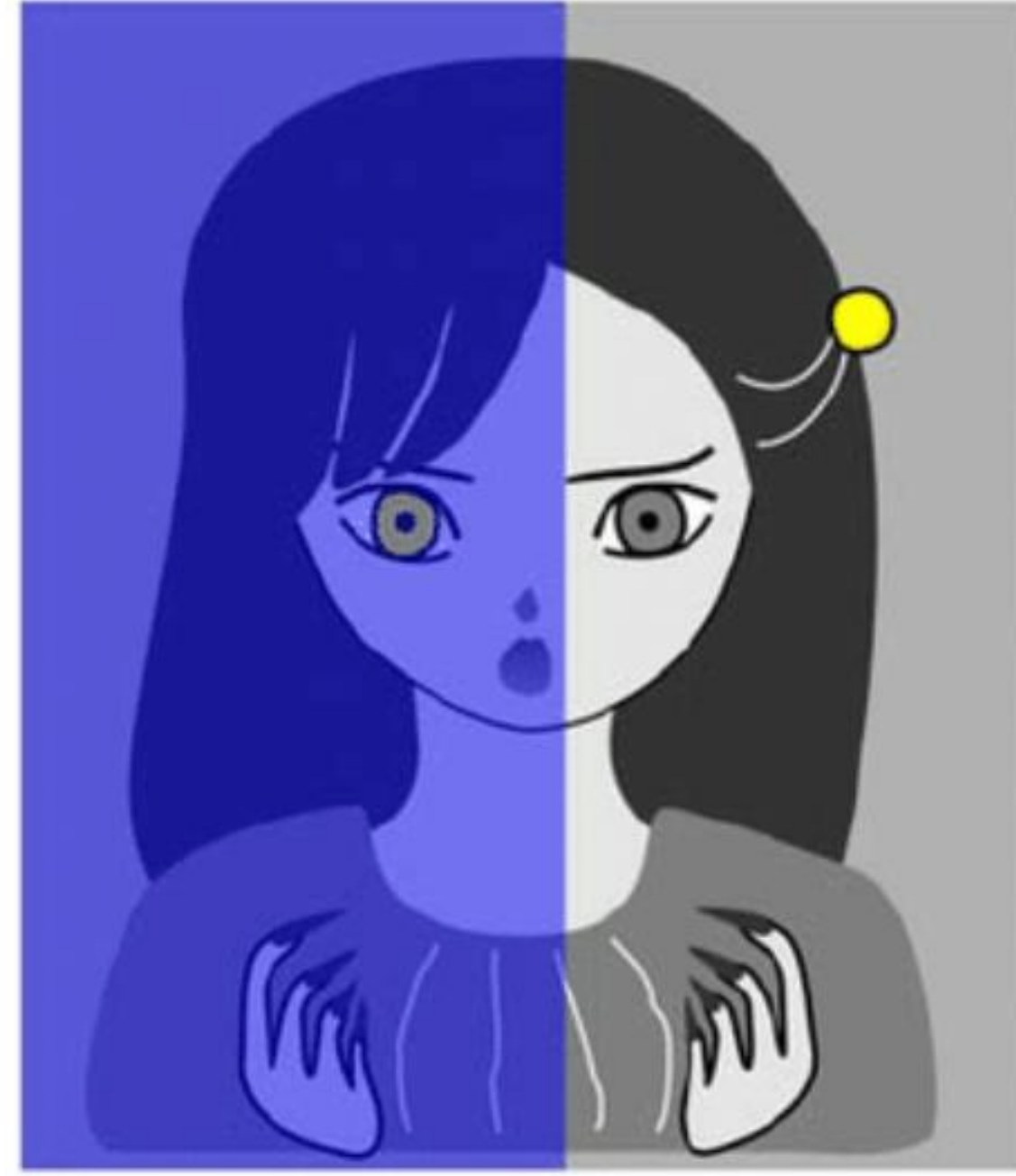
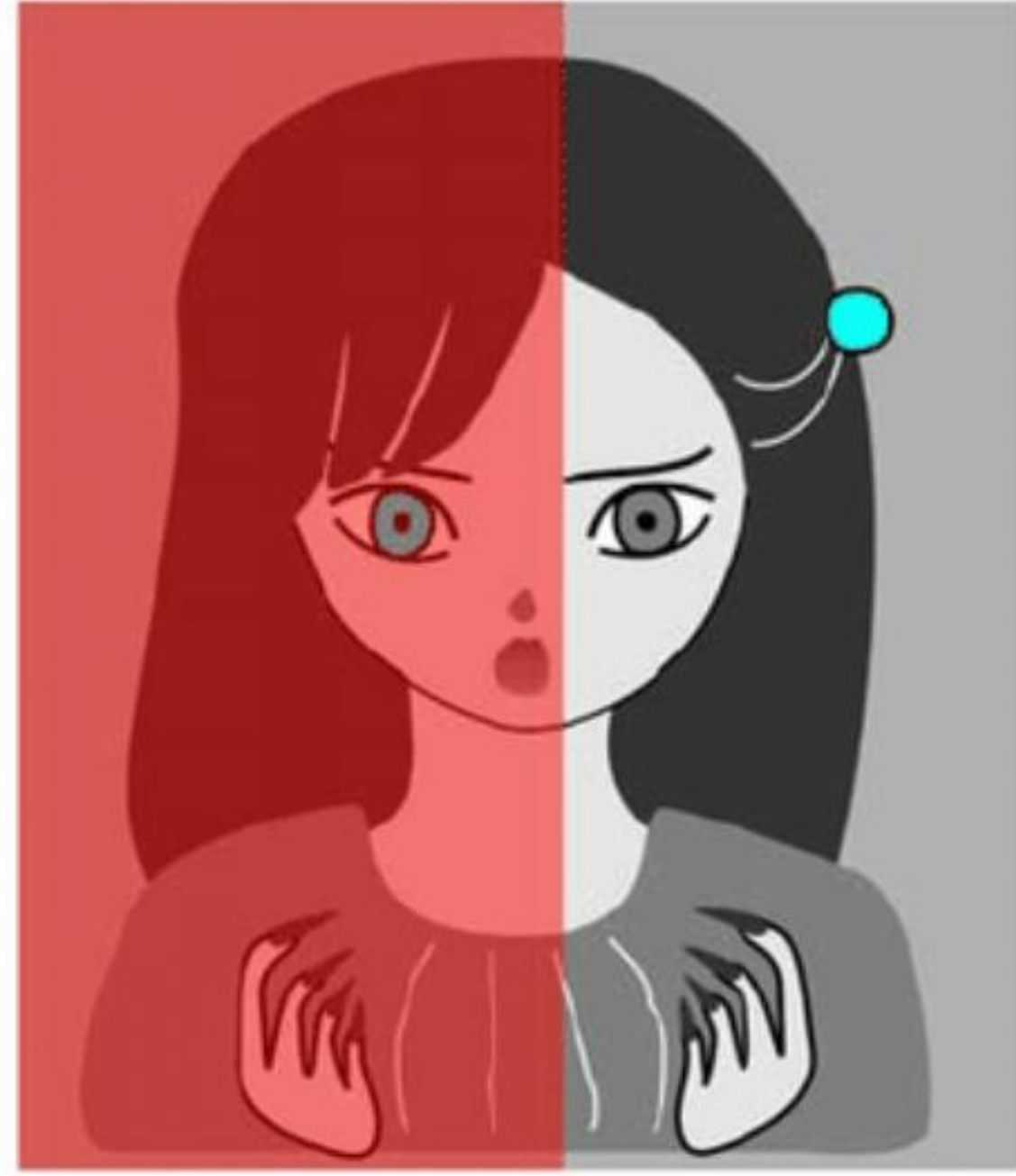
1982

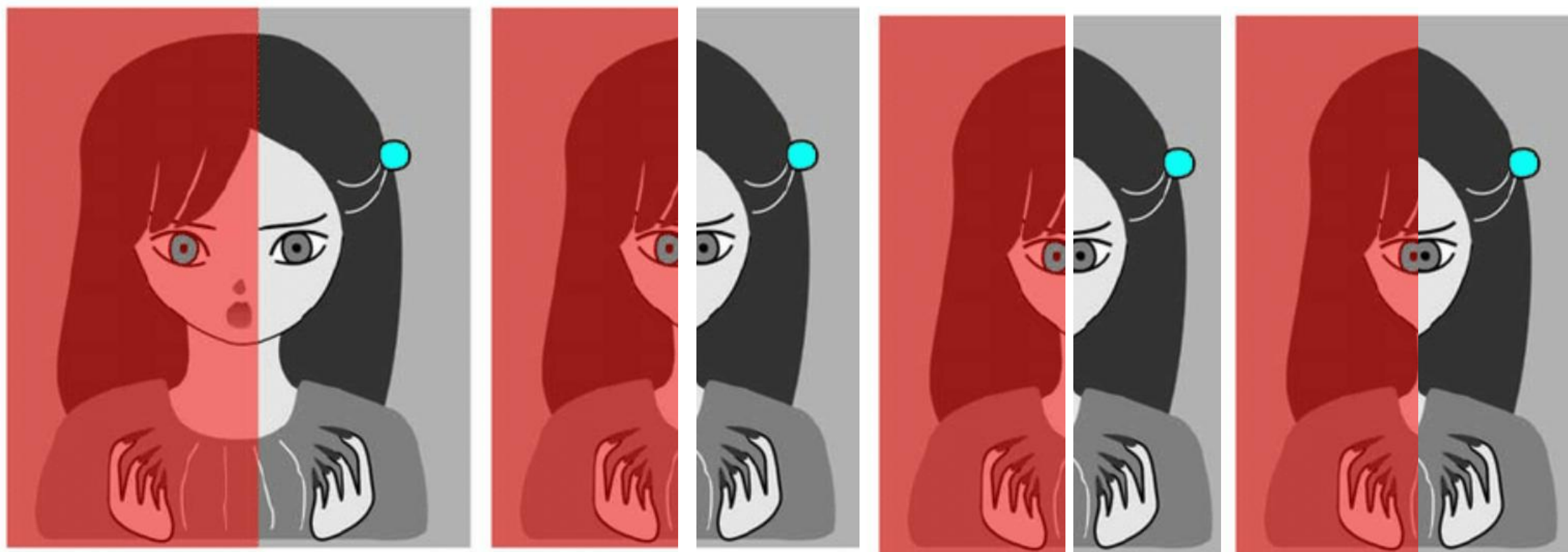
**Publicación de “Visión”
de Marr**

***El cerebro como sistema de
procesamiento de la información:
entenderlo es entenderlo en tres
niveles: computacional,
algorítmico y físico***









Niveles de análisis del cerebro

¿Qué hace?

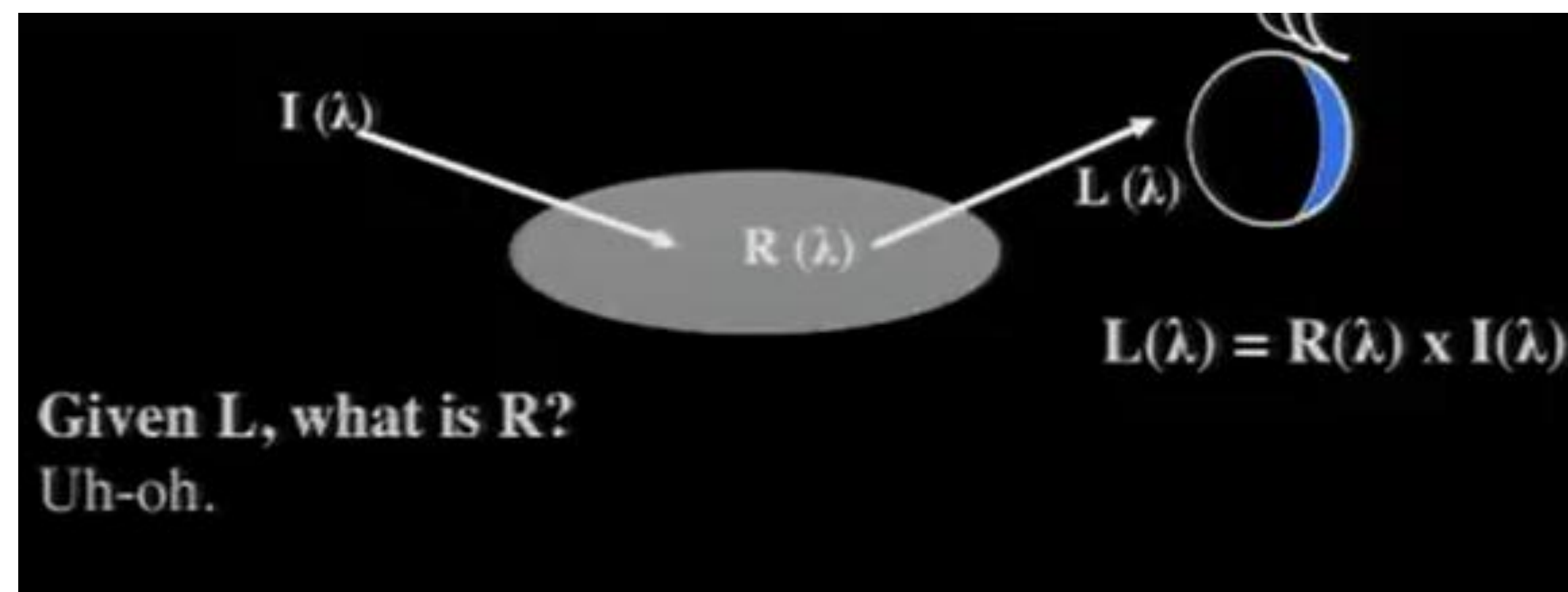
Infiere la reflectancia (R) de los objetos (su color)

¿Cómo lo hace?

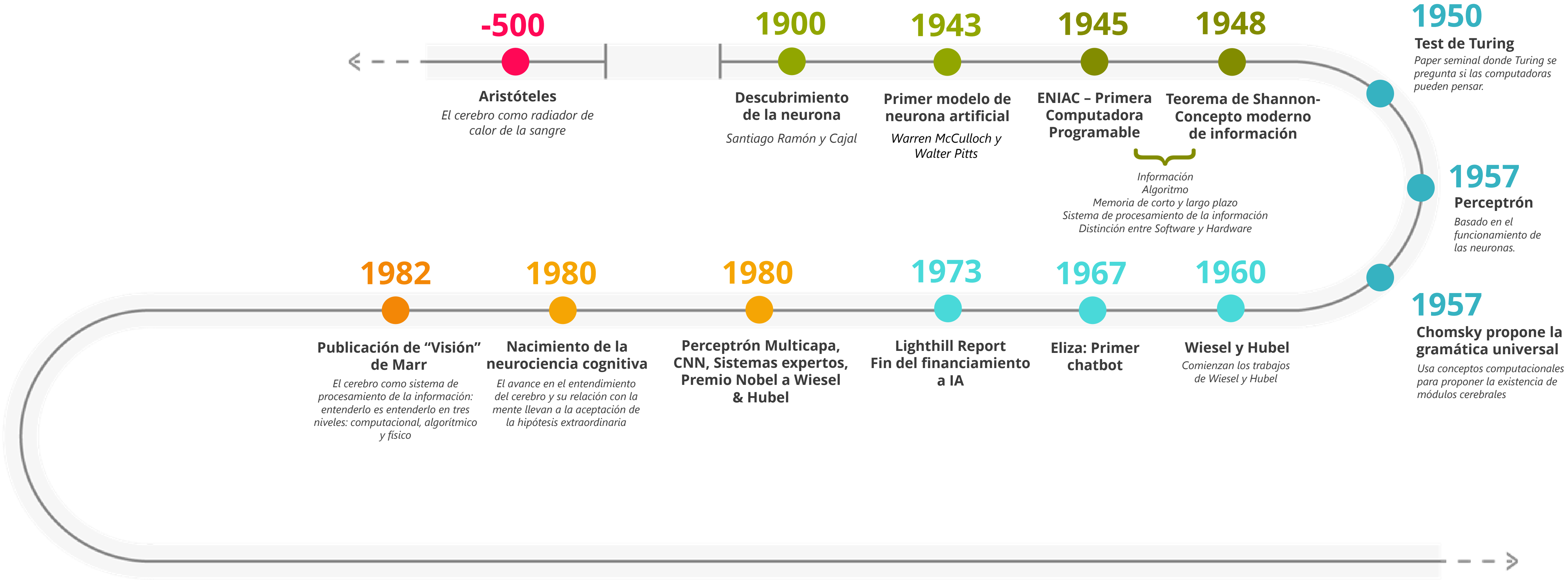
Con receptores de rojo, verde y azul mide (L), por contexto infiere (I) y con eso calcula (R)

¿Cómo es el hardware?

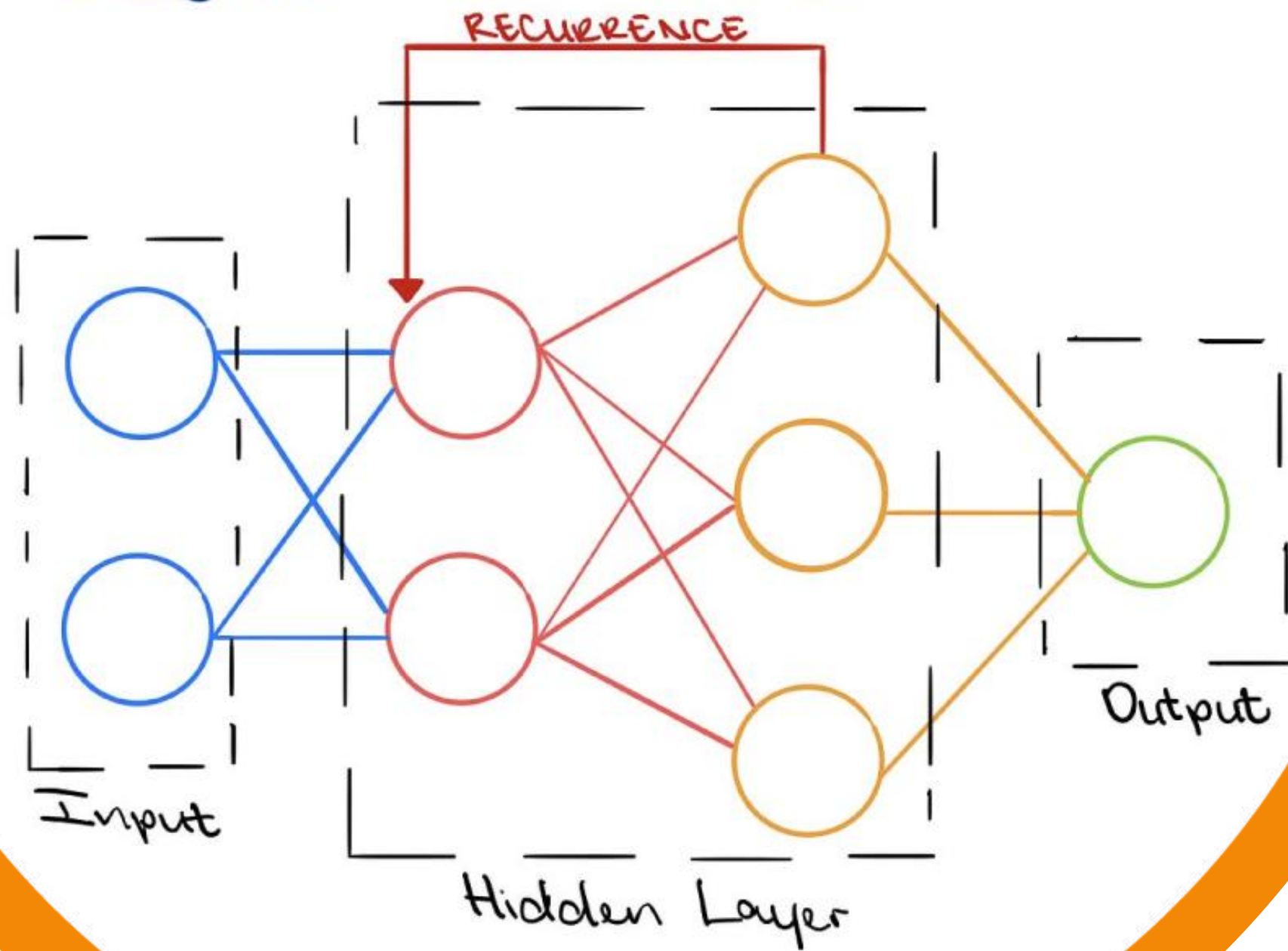
Mediantes células fotoeléctricas llamadas conos y bastones



Neuro e IA: coevolución



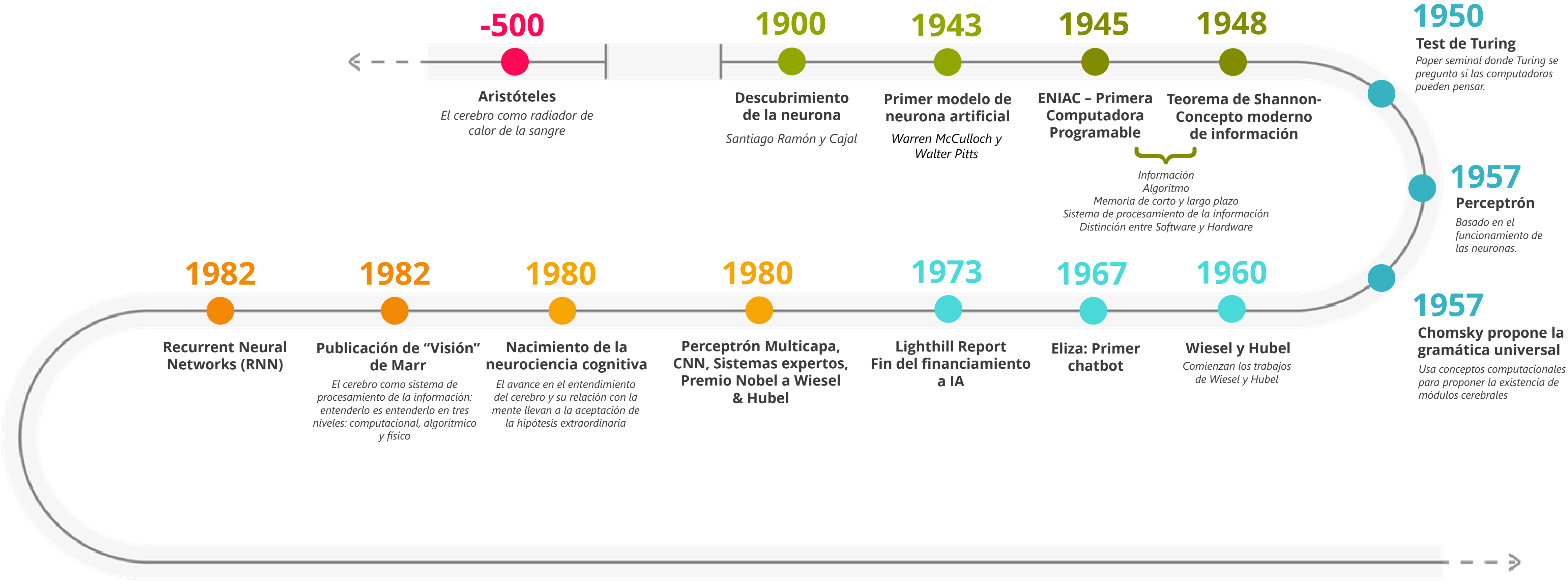
RECURRENT NEURAL NETWORKS



1982

**Recurrent Neural
Networks (RNN)**

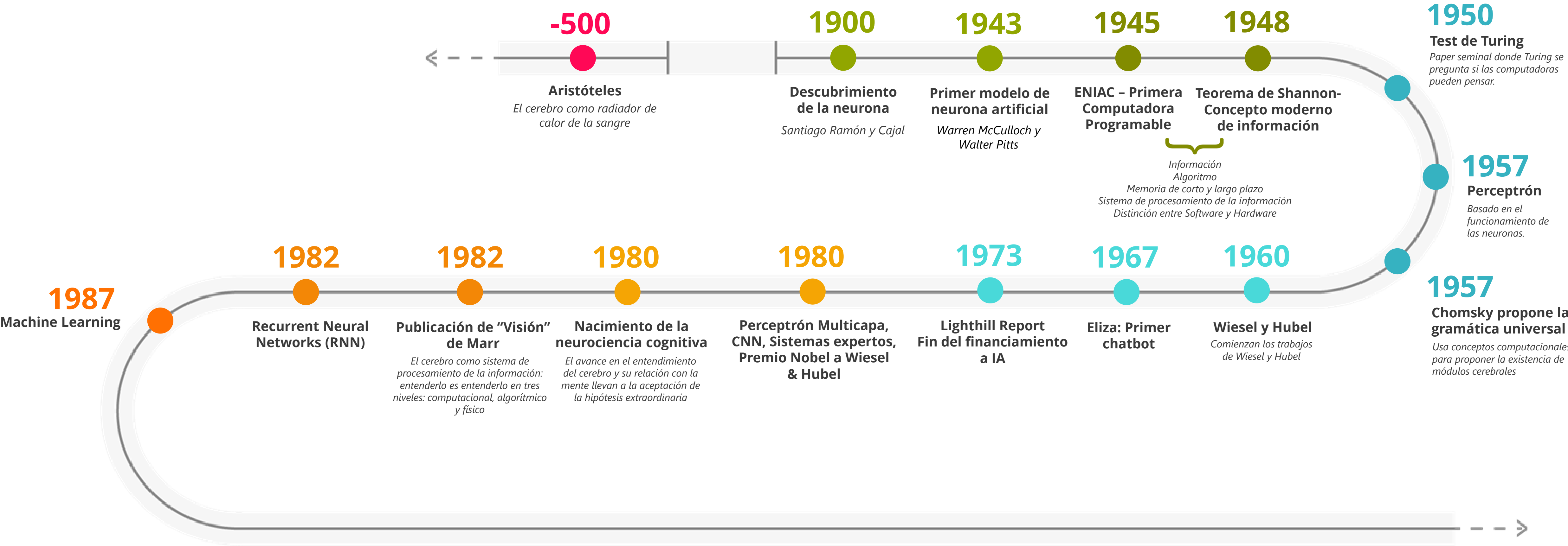
Neuro e IA: coevolución





Nace Machine Learning

Neuro e IA: coevolución

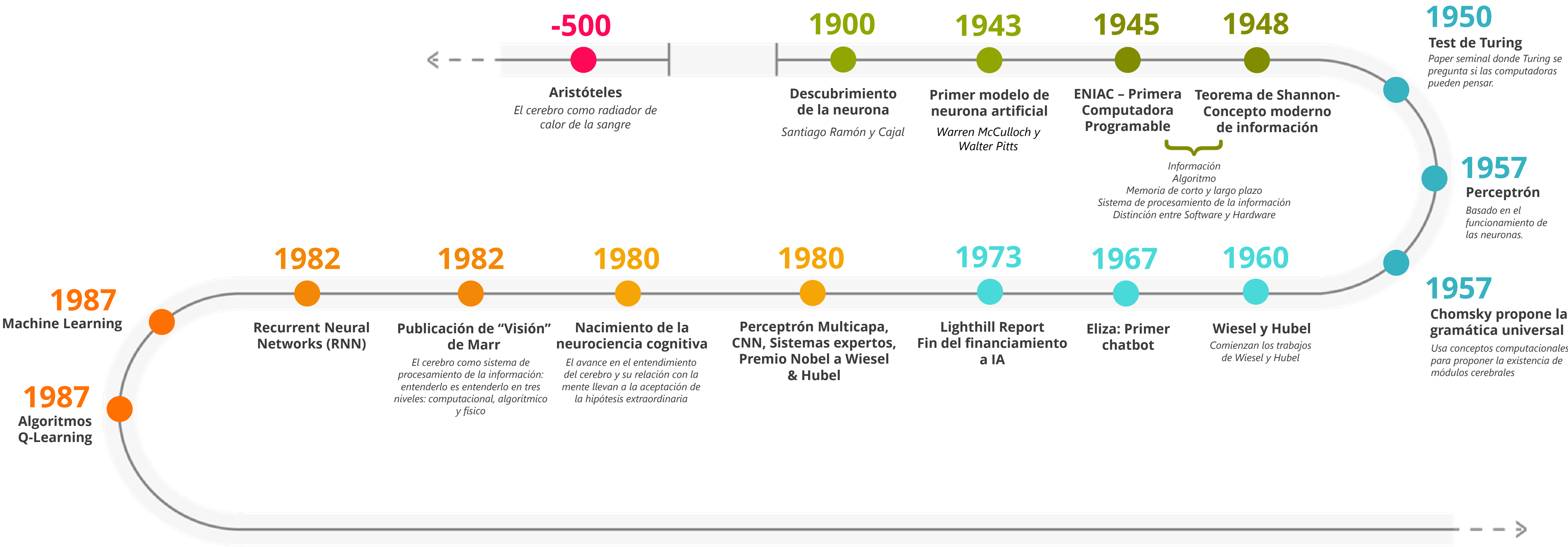




1987

**Algoritmo Q-Learning de
Reinforcement Learning**

Neuro e IA: coevolución

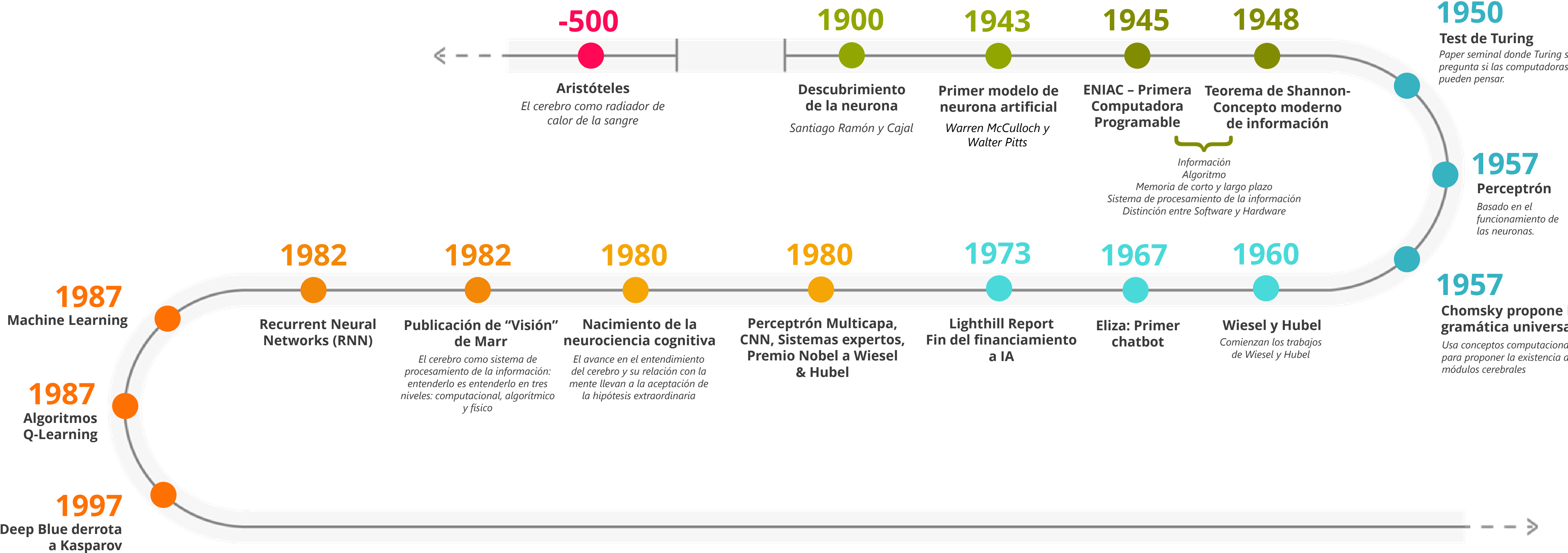




1997

**Deep Blue derrota a
Kasparov**

Neuro e IA: coevolución

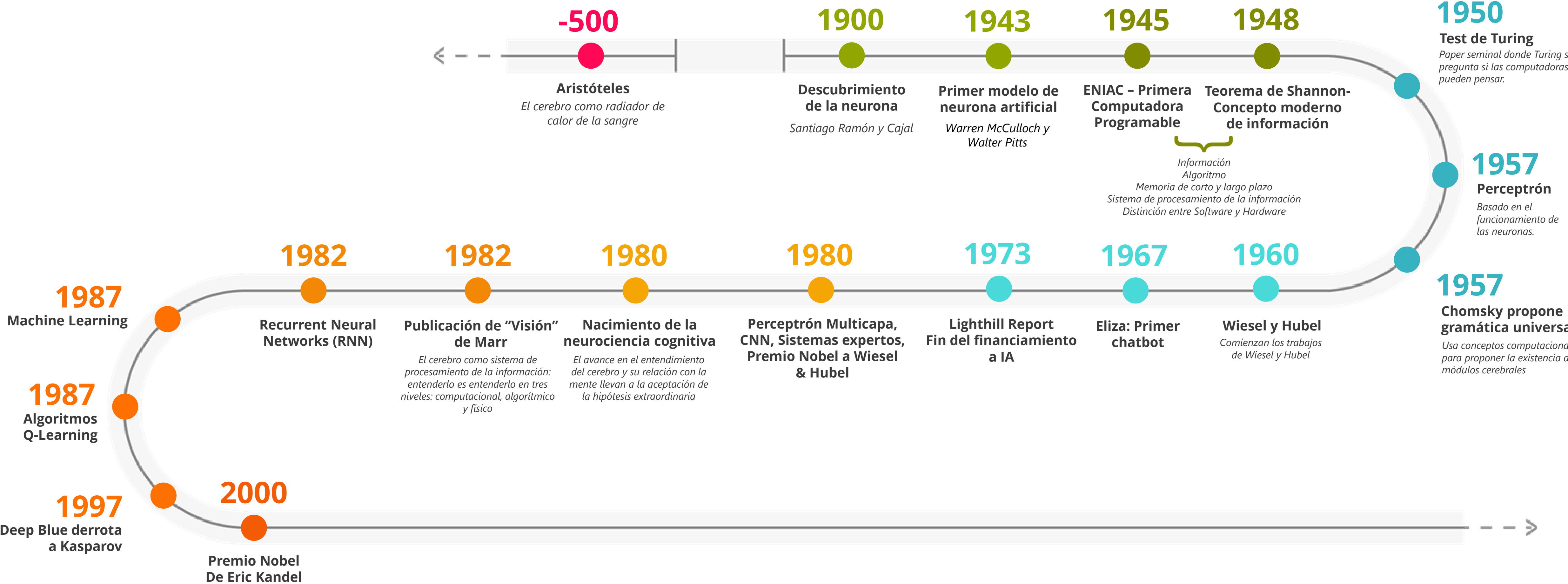




2000

**Premio Nobel
de Eric Kandel**

Neuro e IA: coevolución



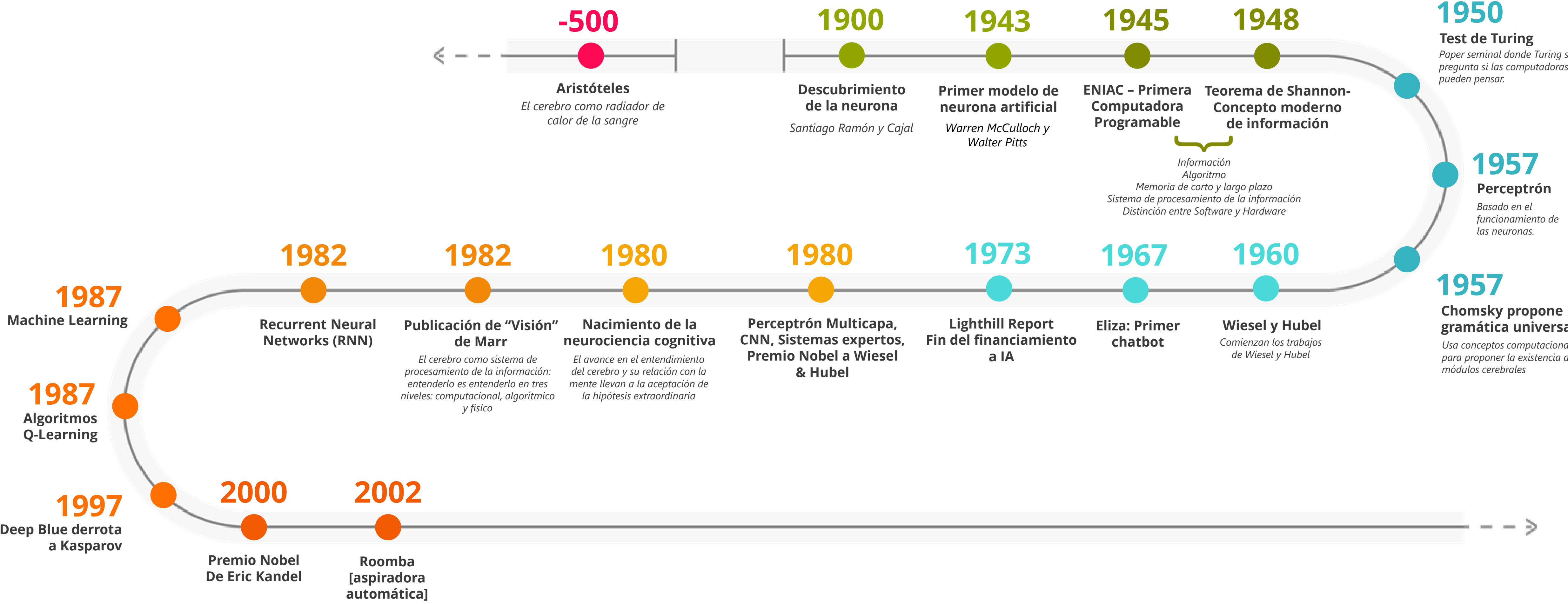


2002

Roomba

Aspiradora automática

Neuro e IA: coevolución

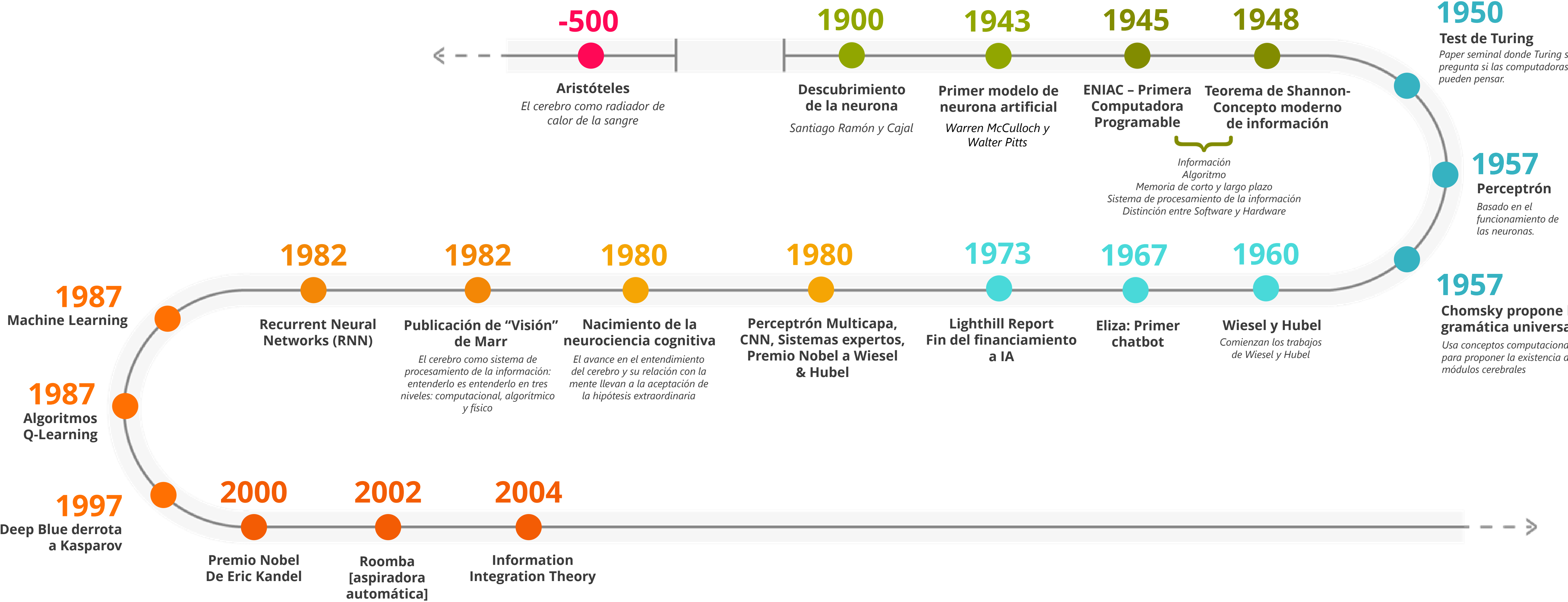




2004

**Information Integration
Theory**

Neuro e IA: coevolución

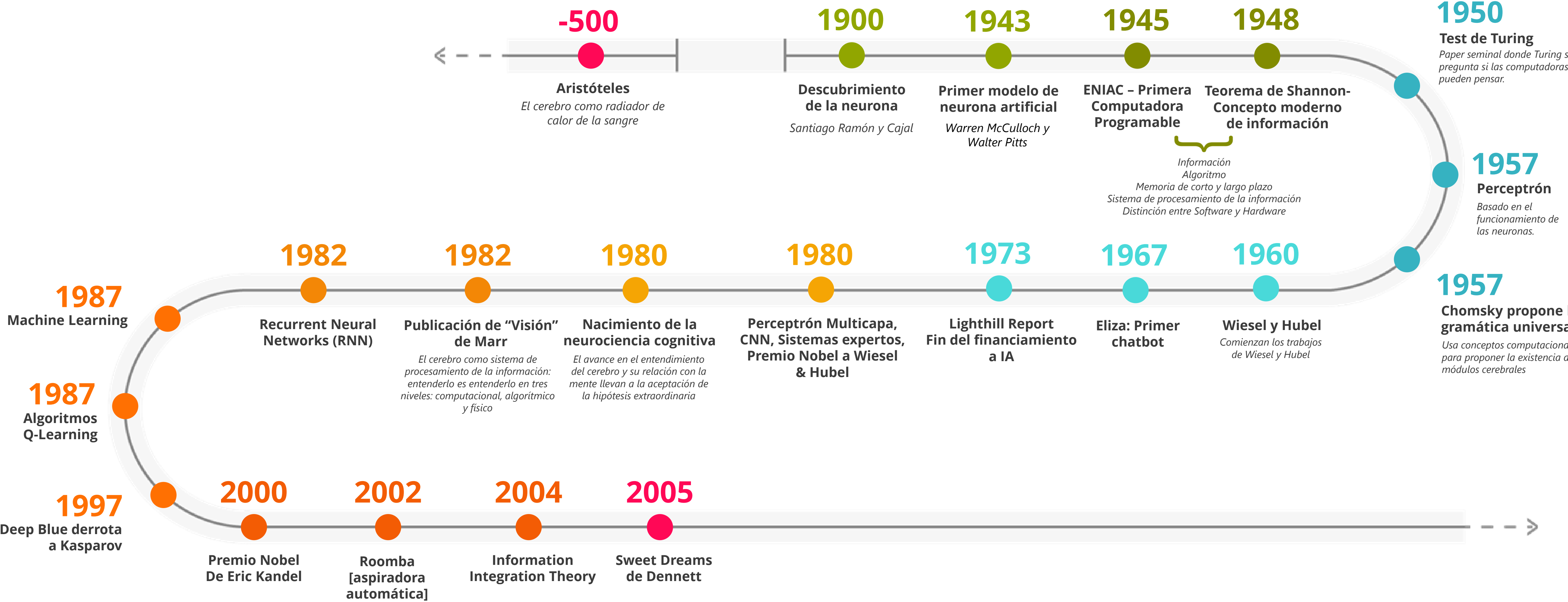


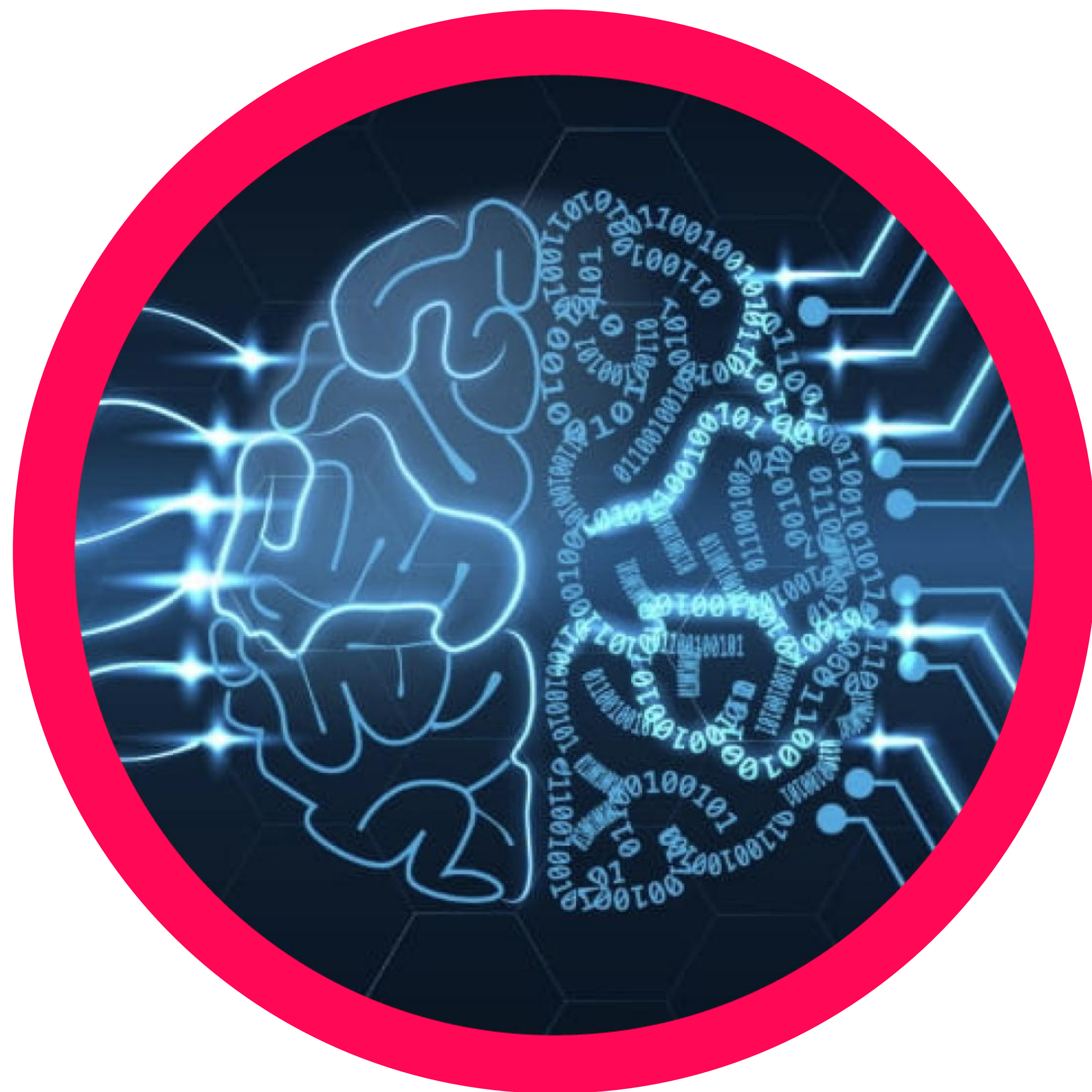


2005

**Publicación de Sweet
Dreams de Dennett**

Neuro e IA: coevolución

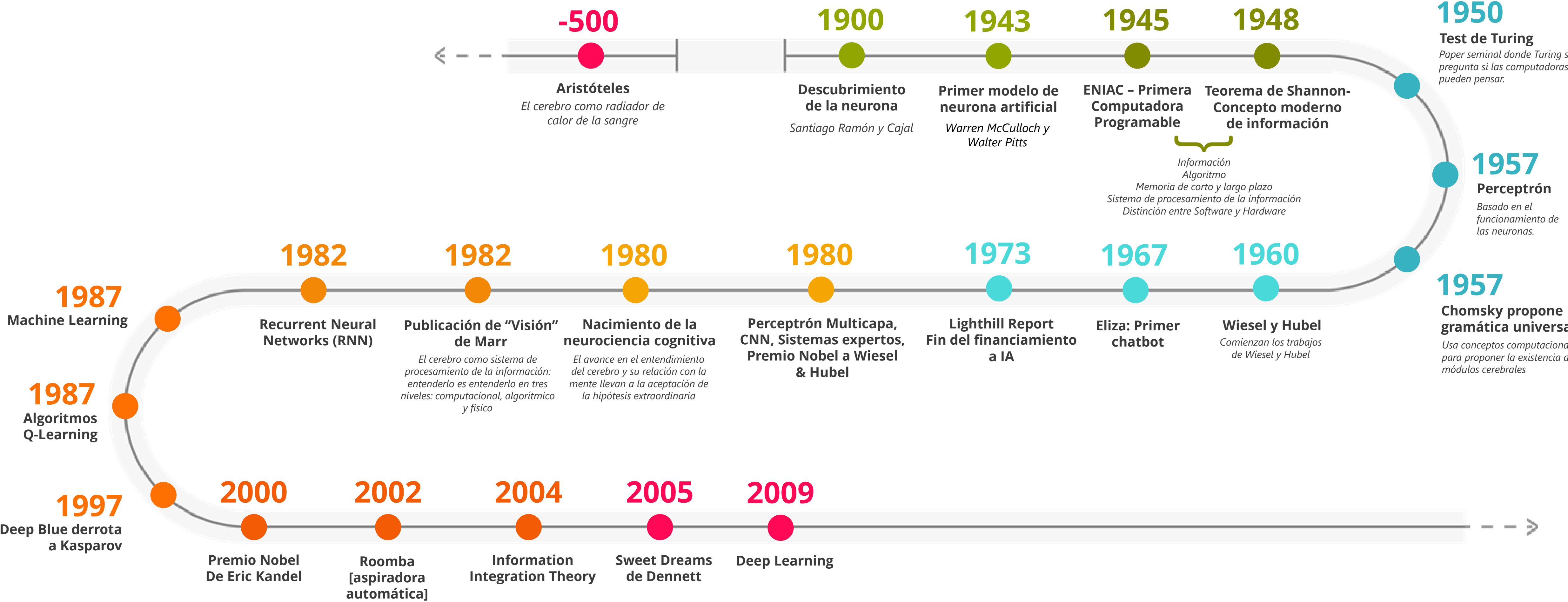




2009

Surgimiento de Deep Learning

Neuro e IA: coevolución

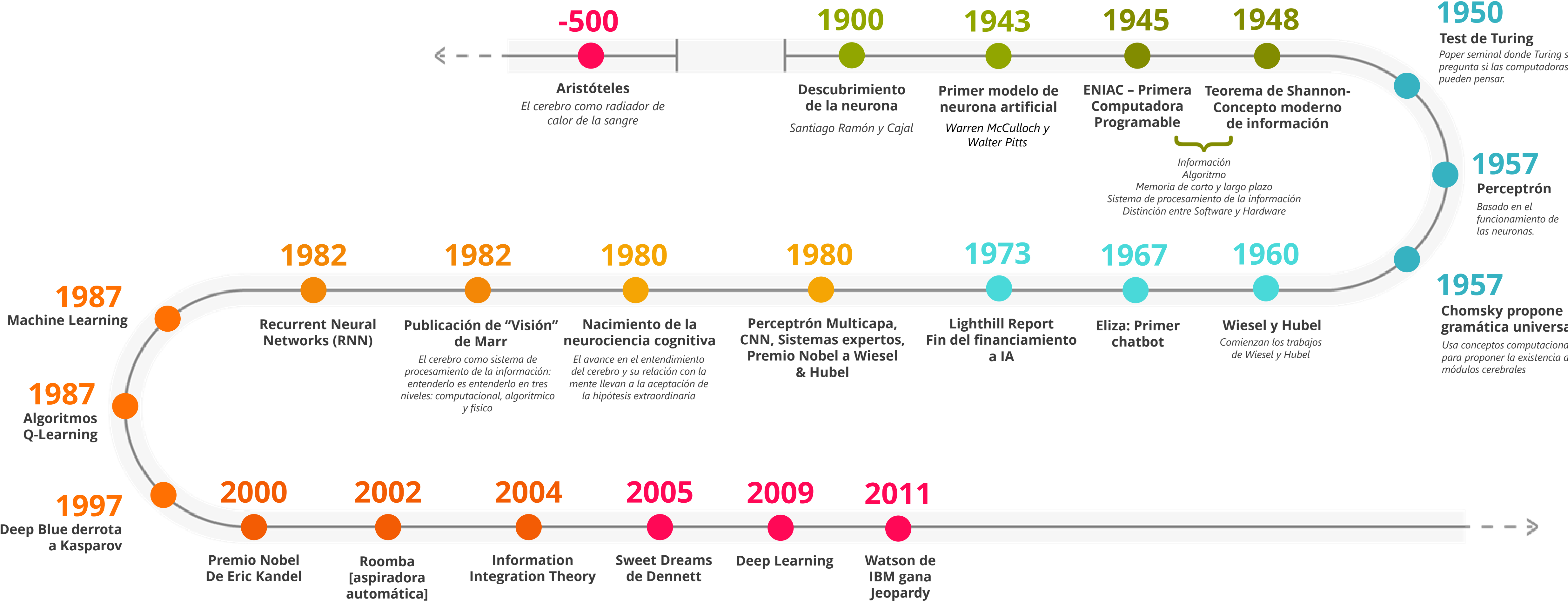




2011

**Watson de IBM
gana Jeopardy**

Neuro e IA: coevolución

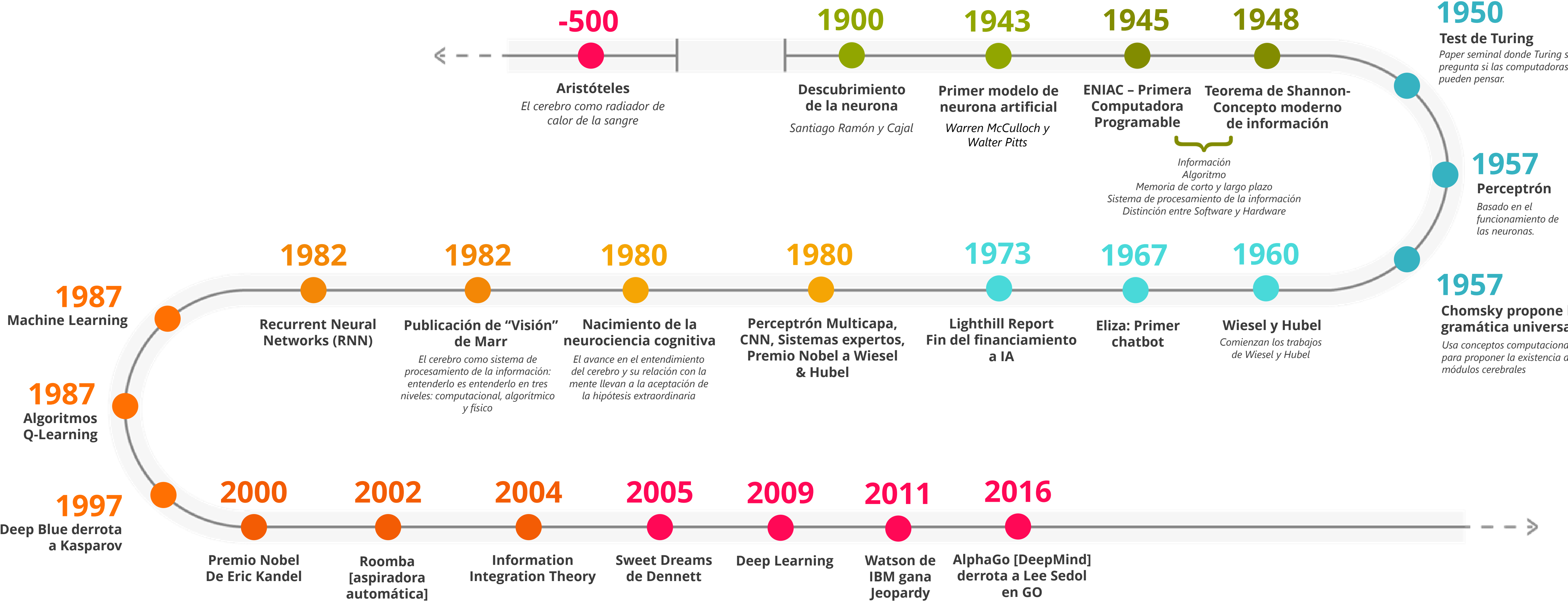




2016

**AlphaGo [DeepMind]
derrota a Lee Sedol en GO**

Neuro e IA: coevolución

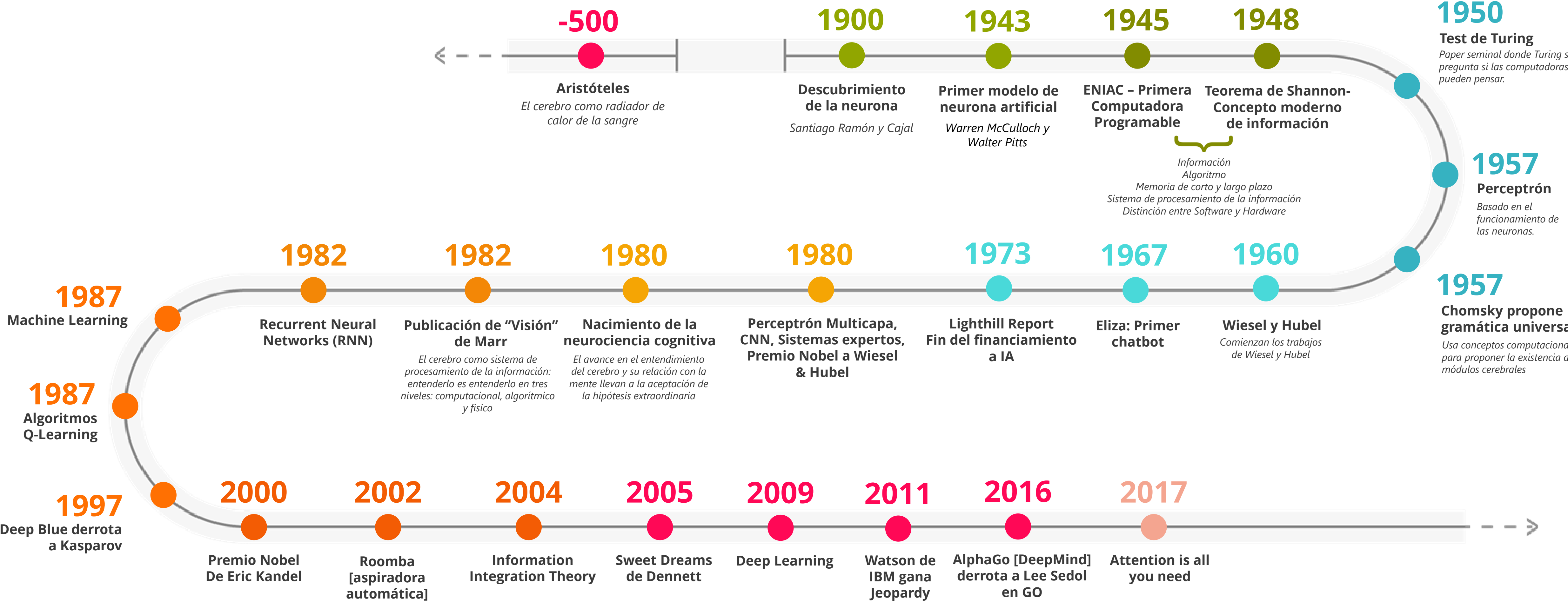




2017

Attention is all you need

Neuro e IA: coevolución



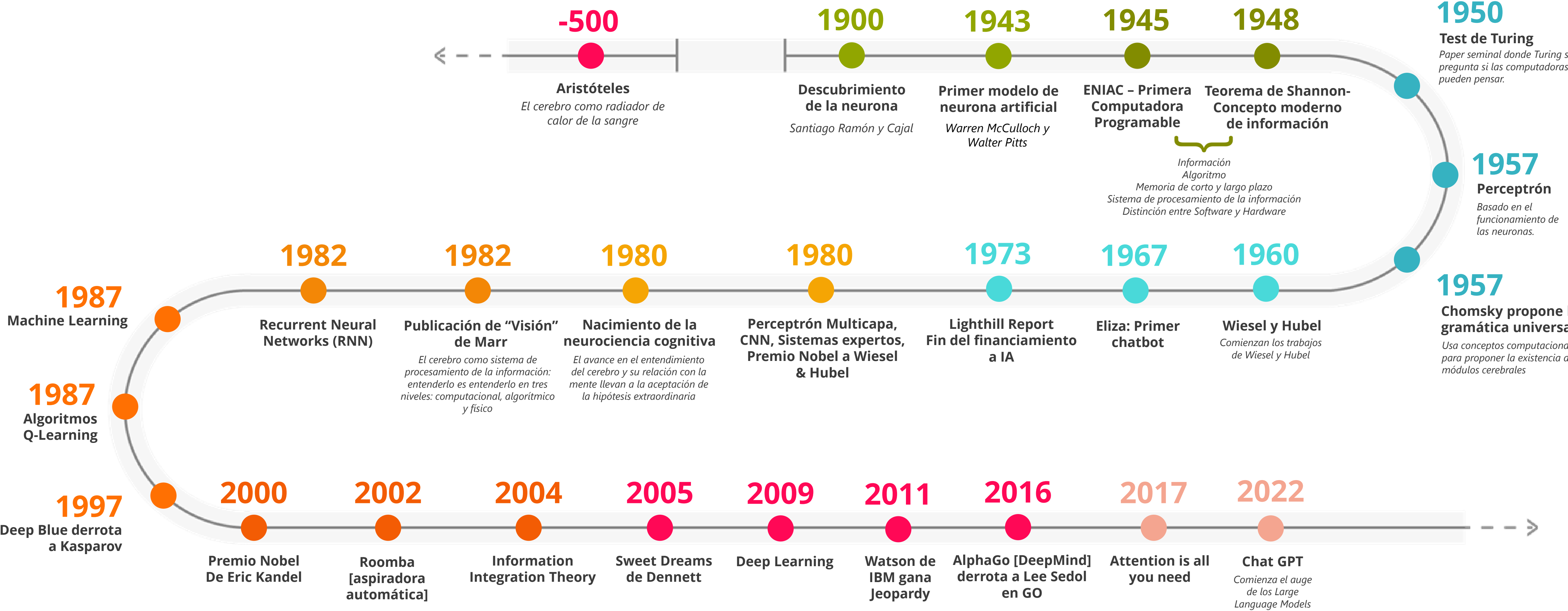


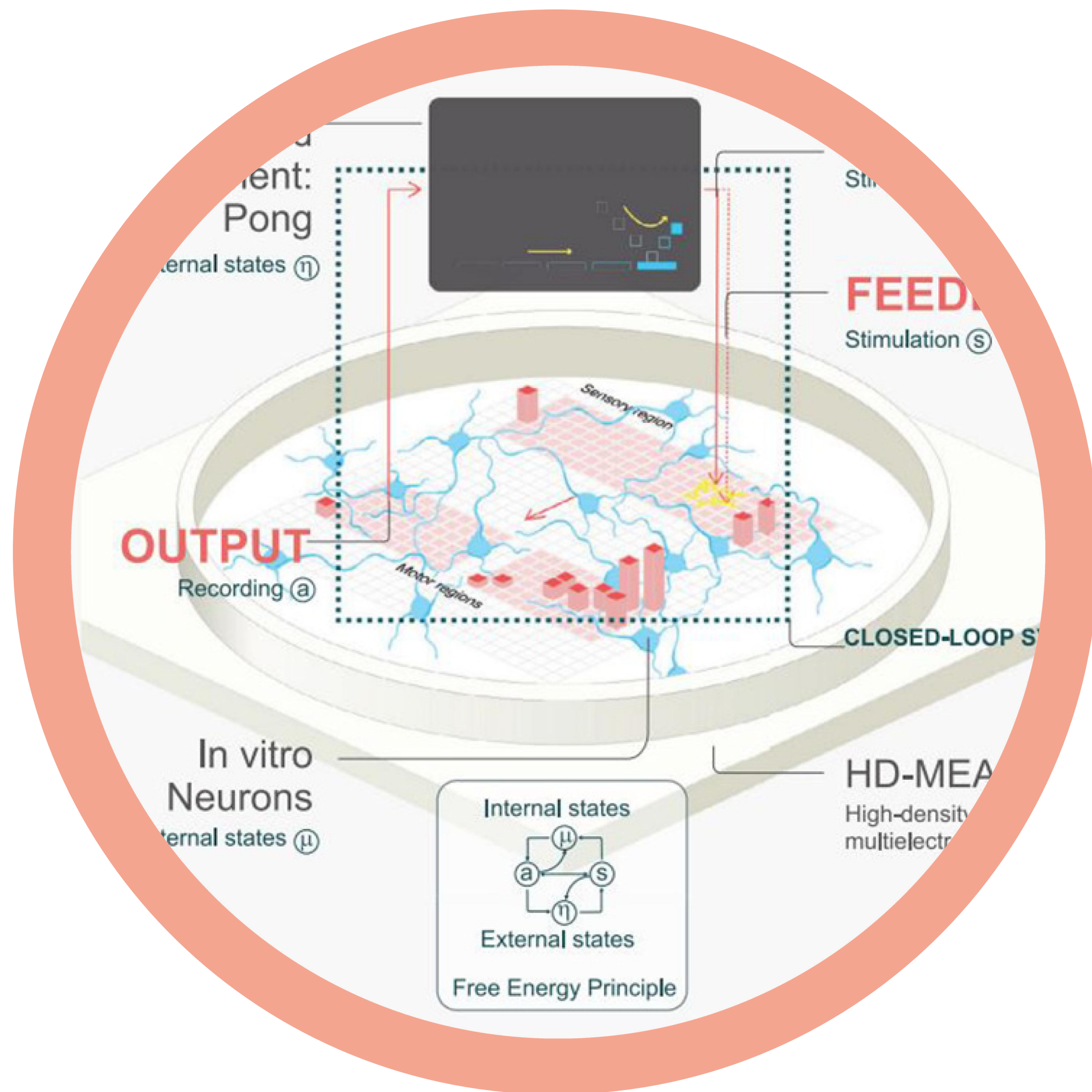
2022

Chat GPT

*Comienza el auge de los
Large Language Models*

Neuro e IA: coevolución





2022

DishBrain

Neuronas humanas en un plato aprender a jugar al Pong

Neuro e IA: coevolución

